



ADVANCING DIGITAL PAYMENTS IN NEPAL

INFRASTRUCTURE UPGRADES AND POLICY
DEVELOPMENT FOR ENHANCED TRADE FACILITATION

JULY 2026

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6 ADB Avenue, Mandaluyong City, 1550 Metro Manila, Philippines
Tel +63 2 8632 4444; Fax +63 2 8636 2444
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Abbreviations

ADB	Asian Development Bank
AML	anti-money laundering
API	application programming interface
BIMSTEC	Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation
CBDC	Central Bank Digital Currency
CFT	combating the financing of terrorism
COVID-19	coronavirus disease
DOCSCP	Department of Commerce, Supplies, and Consumer Protection
DPI	digital public infrastructure
FATF	Financial Action Task Force
GDP	gross domestic product
GNI	gross national income
ISO	International Organization for Standardization
KYC	know your customer
MNO	mobile network operator
MOCIT	Ministry of Communications and Information Technology
MOF	Ministry of Finance
MOICS	Ministry of Industry, Commerce, and Supplies
MSMEs	micro, small, and medium-sized enterprises
NBSM	Nepal Bureau of Standards and Metrology
NCHL	Nepal Clearing House Limited
NPCI	National Payments Corporation of India
NRB	Nepal Rastra Bank
PSO	payment system operator
PSP	payment service provider
QR	Quick Response
SASEC	South Asia Subregional Economic Cooperation
SMS	short message service
UPI	Unified Payments Interface
USSD	Unstructured Supplementary Service Data

Executive Summary

This report has been produced as part of the South Asia Subregional Economic Cooperation (SASEC) program, with support from the Asian Development Bank, to strengthen cross-border trade. Digital payment is globally recognized as a key enabler of cross-border trade, facilitating faster, more secure transactions, and expanding opportunities for businesses to participate in regional and global markets.

Nepal's 16th Five-Year Periodic Plan 2024–2029 identifies digital transformation—implemented through the Digital Nepal Framework—as a key strategy to enhance economic competitiveness and productivity. The Plan aims to expand internet access to 90% of the population by 2029 and transition all public sector payments to digital formats. Nepal is making a concerted push to build a digital public infrastructure (DPI) that will serve as the backbone for its digital economy. Elaborated in the forthcoming Digital Nepal Framework 2.0, the government envisions a unified architecture comprising interoperable digital payment systems, digital national identification, and secure data exchange, designed to enable economic transformation and inclusive growth across sectors, and support cross-border trade.

This report has been developed in alignment with the goals of Nepal's 16th Five-Year Periodic Plan and the Digital Nepal Framework, and supports the country's commitment to building secure and interoperable digital payment systems as part of its DPI. The report assesses Nepal's digital payment ecosystem, a complex network of stakeholders, technologies, processes, and regulations that facilitate the exchange of monetary value for goods and services, domestically and internationally. The report identifies key challenges affecting the adoption and growth of digital payment in Nepal. Subsequently, drawing on lessons from other countries' experiences, the report offers strategic recommendations to strengthen Nepal's digital payment ecosystem.

Key Findings

Nepal, through a program of work under the leadership of Nepal Rastra Bank (NRB), has made significant progress in advancing the digital payment ecosystem, rapidly expanding adoption of mobile and internet banking, and Quick Response (QR) payments among its population. Integration efforts with India's Unified Payments Interface have enabled cross-border payments, demonstrating early success in regional digital payment connectivity. These developments mark a major step forward in digitalizing Nepal's payment infrastructure and laying the foundation for cross-border digital trade.

A feature of Nepal’s digital payment landscape is the presence of multiple payment system operators with distinct systems that payment service providers (PSPs) need to individually connect to. This has resulted in varying levels of interoperability, often requiring merchants and consumers to use different apps or platforms depending on the service. In recognition of this challenge, NRB published the National Payment Switch and the National Payment Ecosystem Master Reference Document 2025, which lays out a blueprint for interoperability standards and shared infrastructure. This initiative signals an important shift from disparate systems toward a shared DPI model to support seamless domestic and cross-border payment flows.

Digital payments have served as an entry point for innovation and growth in digital financial services in Nepal, such as digital lending to increase access to finance for micro, small, and medium-sized enterprises (MSMEs), and the exploration of a Central Bank Digital Currency to facilitate cross-border trade. The Government of Nepal is also developing the digital national identification system and promoting its integration into public services and payment platforms for know-your-customer (KYC) verification.

Digital payment adoption remains uneven among individuals and MSMEs. While the use of digital payment systems has grown in recent years, rural populations and smaller businesses have progressed at a different pace due to varying levels of connectivity and digital and financial literacy. To support more inclusive uptake, some PSPs have introduced offline payment innovations, demonstrating how technological innovation can extend the reach of digital payments in areas with unreliable connectivity.

Continued innovation in payment processing systems and infrastructure will require investments in nurturing a skilled, future-ready workforce—developing talent in multidisciplinary areas such as digital finance, payments technology, cybersecurity, data governance and protection, and interoperability. Beyond technical expertise, strategic leadership and policy capacity are needed in both the public and private sectors to guide innovation and manage emerging risks. Strengthening these capacities will be critical for unlocking new economic opportunities, including in tourism and remittances.

Key Recommendations

To strengthen Nepal’s digital payment ecosystem and enhance its role in trade facilitation, a set of recommendations has been developed. They are organized into three interrelated areas: (i) trade facilitation priorities, which focus on strengthening cross-border payment infrastructure, regulatory alignment, payment system interoperability, and market integration; (ii) broader digital payment system development, which outlines measures to enhance the domestic infrastructure and capacity; and (iii) an implementation framework that emphasizes coordinated governance and stakeholder engagement. The report outlines a multiyear road map to guide the phased implementation of these recommendations.

Trade Facilitation Priorities

1. Cross-Border Payment Infrastructure

- Expedite the resolution of commission structure issues for Nepal–India QR payments to enable the already-developed reciprocal functionality for Nepalis in India.
- Implement dedicated payment corridors with key trading partners featuring enhanced reliability and pursuing regional standardization through SASEC.

- Deploy specialized payment solutions for the economically vital tourism and remittance sectors with multicurrency capabilities and streamlined onboarding.
- Enhance reliability of international payment gateways by addressing middleware issues and establishing redundant infrastructure with automatic failover capabilities.

2. Regulatory Alignment and Compliance

- Strengthen anti-money laundering/combating the financing of terrorism frameworks to address Financial Action Task Force requirements.
- Implement risk-based KYC procedures that balance financial inclusion with appropriate safeguards, building on the existing tiered model.
- Harmonize foreign exchange regulations with trading partners and address challenges posed by Nepal's currency peg with the Indian rupee.
- Develop and enact a comprehensive Personal Data Protection Act aligned with international standards.
- Establish clear consumer protection regulations for digital financial services, especially for cross-border transactions.
- Create a regulatory framework for e-commerce transactions.

3. Payment System Interoperability

- Finalize and mandate the planned open application programming interface rule book to create standardized interfaces connecting banks, businesses, and government agencies.
- Drive market-wide adoption of the unified NEPALPAY QR standard to address current fragmentation and enable cross-border QR payments with key trading partners.
- Establish interoperability between mobile wallets and banking systems to improve routing and reduce transaction costs.

4. Trade-Focused Market Development

- Introduce incentive programs for export-oriented businesses.
- Develop specialized digital solutions for informal sector integration.
- Implement digital documentation solutions for cross-border trade.
- Establish cross-border fintech partnerships between Nepali PSPs and international fintech companies to adapt proven global solutions to Nepal's needs.

Broader Digital Payment System Development

5. Digital Infrastructure Enhancement

- Deploy low-bandwidth and offline-capable payment solutions for areas with limited connectivity.
- Implement hybrid connectivity models combining fiber-optic networks with alternative technologies to reach rural areas.
- Support the effective implementation of the Cyber Resilience Guidelines and establish a national Financial Sector-Computer Security Incident Response Team.
- Establish monitoring systems with real-time visibility and automated alerts to track and improve system performance.

6. Market Development and Capacity Initiatives

- Launch targeted merchant adoption programs with subsidized equipment, fee waivers, and training resources, particularly for MSMEs.
- Develop a nationwide network of cash-in/cash-out agents to ensure last-mile financial inclusion.
- Create a public–private partnership framework to incentivize private sector investment in expanding access to digital financial services and agent networks in underserved regions.
- Develop comprehensive digital and financial literacy programs with specialized modules for different user groups, including hands-on training and multilingual materials.
- Address skill shortages through partnerships with educational and training institutions to develop specialized fintech curriculum and certification programs.
- Create a formal regulatory sandbox, as planned by NRB, to foster controlled innovation.
- Clarify regulatory rules for different providers of digital payments, and ensure fair infrastructure access for licensed nonbank PSPs.

Implementation Framework

The analysis emphasizes the need for careful sequencing of initiatives, with infrastructure reliability and regulatory clarity as foundational requirements. Achieving success requires a balanced attention to both trade facilitation and domestic system development. Implementation needs to be guided by regular stakeholder feedback and adaptive management approaches to ensure sustainable progress toward a resilient digital payment ecosystem that supports both domestic and international transactions. Central to this effort is strong coordination among stakeholders, including NRB, financial institutions, technology providers, and the business community.

I. Introduction

Country Context

Nepal is a country in South Asia with a diverse topography encompassing the Himalayan mountains, hills, and the fertile plains of the Terai region. Its population of 29.65 million people¹ is equally diverse, comprising numerous indigenous and linguistic communities.

Nepal's economy is predominantly agrarian, with agriculture, forestry, and fishing employing about 60% of the population and contributing to 21% of the country's gross domestic product (GDP).² However, the service sector has been expanding, driven by tourism and remittances from Nepalese working abroad. In 2023, around 55% of Nepal's GDP came from its service sector.³

Overall, Nepal's economy continues to recover from the coronavirus disease (COVID-19) pandemic, with GDP growth rising from 2.0% in 2023 to 3.9% in 2024, and growth is forecasted to continue, reaching 4.4% in 2025 and 5.1% in 2026.⁴ The tourism sector, severely impacted by the pandemic, reached 96% of pre-pandemic visitor levels by 2024.⁵ Remittances, amounting to NRs1.52 trillion (\$11 billion) in 2023 and accounting for over 26% of the GDP,⁶ have been instrumental in poverty alleviation and sustaining domestic consumption during and after the pandemic.

Regional cooperation and trade linkages are recognized as pivotal for Nepal to mitigate the challenges of its geography and relatively small domestic market.⁷ India is Nepal's largest trading partner, accounting for over 60% of its total trade volume.⁸ The Nepalese rupee is pegged to the Indian rupee at a fixed rate of NRs1.60 to ₹1.00, which has provided monetary stability, but has also reinforced its economic and financial interdependence with India.⁹ Efforts are underway to diversify trade partners, with initiatives to enhance connectivity and trade relations

¹ World Bank. Data: Nepal. <https://data.worldbank.org/country/nepal> (accessed 10 November 2025).

² L. N. Yogi, T. Thalal, and S. Bhandari. 2025. *The Role of Agriculture in Nepal's Economic Development: Challenges, Opportunities, and Pathways for Modernization*. Heliyon. 11 (2). <https://www.sciencedirect.com/science/article/pii/S2405844025002403>.

³ Statista. 2025. *Distribution of Gross Domestic Product (GDP) Across Economic Sectors Nepal 2023*. <https://www.statista.com/statistics/425750/nepal-gdp-distribution-across-economic-sectors/>.

⁴ Asian Development Bank (ADB). 2025. *Asian Development Outlook: Nepal*. <https://www.adb.org/sites/default/files/publication/1044336/nep-ado-april-2025.pdf>.

⁵ S. Pokharel. 2024. Post-COVID-19 tourism: Nepal achieves 96.3 percent recovery. *Nepal Republic Media*. <https://www.myrepublica.nagariknetwork.com/news/post-covid-19-tourism-nepal-achieves-963-percent-recovery-675c6103b4b4a.html>.

⁶ International Organization for Migration. 2024. Nepal on the Right Track to Achieve Cost-effective Remittance Transfers. 14 June. <https://roasiapacific.iom.int/news/nepal-right-track-achieve-cost-effective-remittance-transfers>.

⁷ Government of Nepal, National Planning Commission. 2024. *The Sixteenth Plan (Fiscal Year 2024/25–2028/29: Unofficial Translation*. https://giwmscdnone.gov.np/media/pdf_upload/16TH%20PLAN%20English%20transaltion%202081%20final%20for%20web1_g75lujb.pdf.

⁸ Embassy of India in Kathmandu, Nepal. Commerce Wing Brief. <https://www.indembkathmandu.gov.in/commerce-wing-brief> (accessed 10 November 2025).

⁹ P. Karki. 2022. Pegged currency system and implication in Nepal. *Nepal Economic Forum*. 22 March. <https://nepaleconomicforum.org/pegged-currency-system-and-implication-in-nepal/>.

with the People's Republic of China (its second-largest trade partner) and Bangladesh.¹⁰ Nepal is also a member of regional trade agreements including the South Asian Free Trade Area and the Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation (BIMSTEC).

Nepal is scheduled to graduate from least developed country status in November 2026, which entails gradual phasing out of various international support measures, including preferential trade access. In Nepal's 16th Five-Year Periodic Plan, digital transformation, implemented through its Digital Nepal Framework, is recognized as a key strategy to enhance economic competitiveness and productivity. The 16th Plan targets to expand internet access to 90% of the population by 2029 and "convert all types of payments in public bodies to electronic format" (footnote 7). Development of the finance sector, including digital payment systems, is recognized as one of eight key priorities in the Digital Nepal Framework.

An update of the Digital Nepal Framework has been drafted with plans to submit to the cabinet for approval.¹¹ The draft Digital Nepal Framework 2.0 prioritizes the expansion of digital payment systems, e-commerce, and fintech innovations. It emphasizes the development of a digital public infrastructure (DPI) that integrates digital payments, digital identification, and data exchange systems to expand digital access to services, promote innovation and growth, and transform the economy.¹²

Report Objectives, Structure, and Methodology

This report has been developed in alignment with the goals of the 16th Five-Year Periodic Plan and the Digital Nepal Framework, and supports the country's commitment to building secure and interoperable digital payment systems as part of its DPI.

The report examines the current state of Nepal's digital payment ecosystem, and draws on global best practices and lessons to identify pathways for accelerating its development. It assesses Nepal's digital payment systems, infrastructure, policies, regulations, and market readiness, highlighting key challenges and gaps that affect efficiency, adoption, and cross-border payment integration (Section 2). Section 3 presents country case studies, highlighting the insights and lessons from digital payment implementations across the globe, and their relevance to Nepal's context. Based on this analysis, strategic recommendations to strengthen Nepal's digital payment ecosystem are presented in Section 4.

The report is based on research conducted between October 2024 and October 2025. The analysis combines comprehensive desk research of policy documents, technical documentation, and country case studies with in-depth stakeholder interviews. Interviews were held with senior representatives from Nepal Rastra Bank (NRB), the central bank, and other stakeholders in the digital payment ecosystem (Appendix 1).

¹⁰ Government of Nepal, Ministry of Foreign Affairs. 2025. *Nepal-China Relations*. <https://mofa.gov.np/content/1450/nepal-china-relations/>; I. Hossain. 2025. Cross-Border Cooperation: Nepal-Bangladesh Trade and Its Impact on Local Economy. *The Asian Age*. 3 July. <https://dailyasianage.com/news/339623/cross-border-cooperation-nepal-bangladesh-trade-and-its-impact-on-local-economy>.

¹¹ *Himal Press*. 2025. Second phase of Digital Nepal Framework will begin this fiscal: Gurung. 12 August. <https://en.himalpress.com/second-phase-of-digital-nepal-framework-will-begin-this-fiscal-gurung/>.

¹² Draft Digital Nepal Framework 2.0. https://giwmscdnone.gov.np/media/pdf_upload/dnf_jbji8eb.pdf.

The intended audience for this report includes policymakers at NRB and relevant government ministries, financial institutions, payment system operators (PSOs), payment service providers (PSPs), and development partners involved in Nepal's finance sector. The findings and recommendations aim to inform policy decisions and guide investments in digital payment infrastructure that will support Nepal's trade facilitation objectives.

The report has been produced as part of the South Asia Subregional Economic Cooperation (SASEC) program, with support from the Asian Development Bank (ADB), to strengthen cross-border trade and digital payment infrastructure within the region.

II. The State of Nepal's Digital Payment Ecosystem

The digital payment ecosystem is a complex network of stakeholders, technologies, processes, and regulations that enable the exchange of monetary value for goods and services. Nepal's digital payment ecosystem is shaped by its unique financial, payments, and digital landscape, which is described in sections A, B, and C of this chapter, respectively. This is followed by an assessment of the readiness of businesses in adopting digital payments and other digital tools, especially for trade (Section D). Section E reviews the relevant policies and regulations that enable reliable, secure and trusted digital payments. Based on these findings, a summary of the key challenges identified in Nepal's digital payment ecosystem is presented in Section F.

The Financial Landscape

With NRB as the driving force, and guided by the Nepal Financial Inclusion Action Plan that aims to provide access to affordable finance for all by 2030,¹³ the country has made substantial progress in financial inclusion. The percentage of adults with a transaction account almost doubled from 34% in 2014 to 60% in 2024.¹⁴ There has also been impressive growth in the number of mobile banking users¹⁵—from 1.1 million in 2020¹⁶ to 28.2 million in 2025.¹⁷

The country's banking sector is relatively competitive with a bank concentration ratio of 17.9%,¹⁸ which is among the lowest in the region. This competitive environment has spurred innovation in financial and payment services. The entry of nonbank service providers has further accelerated financial inclusion with mobile wallet services¹⁹ expansion. The number of mobile wallet users grew from 6.2 million in 2020 (footnote 16) to 24.7 million in 2025 (footnote 17).

¹³ NRB. 2017. *Nepal Financial Inclusion Action Plan*. https://un.org.np/sites/default/files/doc_publication/2018-08/MAP%20Nepal%20Financial%20Inclusion%20Action%20Plan.pdf.

¹⁴ This refers to the percentage of adults (age 15+) who have an account at a bank or similar financial institution, or who used a mobile money service in the past year. Source: World Bank. *Inclusive Digital Financial Services: Nepal*. <https://digitalfinance.worldbank.org/country/nepal> (accessed 10 November 2025).

¹⁵ Mobile banking refers to bank-led digital channels that provide users with direct access to their bank accounts for transactions and financial services. The numbers do not necessarily represent unique users.

¹⁶ NRB. 2020. *Monthly Payment Systems Indicators for the Month of: 2077 Shrawan*. 11 September. https://www.nrb.org.np/contents/uploads/2020/09/2077_04_Payment-Systems-Indicators.pdf.

¹⁷ NRB. 2025. *Monthly Payment Systems Indicators: Bhadau, 2082 (Mid-September 2025)*. 15 October. https://www.nrb.org.np/contents/uploads/2025/10/2082_05_PS-Updated-Format.pdf.

¹⁸ Bank concentration refers to the share of a country's total banking assets held by its three largest commercial banks. Source: Trading Economics. *Nepal – Bank Concentration*. <https://tradingeconomics.com/nepal/bank-concentration-percent-wb-data.html> (accessed 10 November 2025).

¹⁹ Mobile wallets are nonbank digital platforms offered by nonbank entities that allow users to store electronic value and make payments, transfers, and purchases without a bank account. The numbers do not necessarily represent unique users.

A national survey of financial service usage in Nepal found 90% of adults using various formal financial services in 2022, including bank accounts, savings, credit, payments, and insurance. Notably, the number of adults who use informal channels decreased from 21% in 2014 to 4% in 2022. The number of adults who do not use any type of formal or informal financial services (the excluded) also decreased from 18% in 2014 to 7% in 2022.²⁰

Nepal has achieved gender parity in account ownership, with about 60% of both women and men having a transaction account in 2024.²¹ However, differences remain in the usage of financial services. For example, only 22% of women made digital payments compared with 35% of men in 2024.²² Similarly, across both rural and urban populations, the use of formal financial services has increased. While uptake continues to grow in rural areas, it remains higher in urban centers, with 77% of adults using financial services in rural areas compared with 84% in urban areas (footnote 20).

In many cases, financial service usage is largely transactional. For instance, beneficiaries are required to open bank accounts to receive government social benefits, but funds are often withdrawn immediately, leaving accounts largely dormant. Similarly, remittances received through bank accounts are typically cashed out rather than retained or used for digital transactions (footnote 20).

Nevertheless, Nepal's large remittance economy is a key driver of financial inclusion and digital payment adoption.²³ The large inflow of remittances has driven the development of specialized financial services. For example, IME Pay²⁴ provided remittance-backed loans specifically for migrant workers and disbursed NRs1.2 billion (\$8.64 million) in loans between 2023 and 2024.²⁵

The uptake of formal credit is increasing, driven by rising entrepreneurship and business expansion, as well as government interventions that offer concessional and subsidized lending to various target groups such as women, returnee migrants, entrepreneurs, and farmers.²⁶ In addition, banks are gradually offering pre-approved, collateral-free credit facilities based on earnings and cash flow, as collateral requirement remains one of the major bottlenecks to accessing formal credit (footnote 20).

²⁰ International Finance Corporation. 2023. Nepal Financial Inclusion Report 2023. <https://www.ifc.org/content/dam/ifc/doc/2023-delta/2023-nepal-financial-inclusion-report.pdf>.

²¹ This refers to the percentage of adults (age 15+) who have an account at a bank or similar financial institution, or who used a mobile money service in the past year. Source: World Bank. Inclusive Digital Financial Services: Nepal. <https://digitalfinance.worldbank.org/country/nepal> (accessed 10 November 2025).

²² This refers to the percentage of adults (age 15+) who used mobile money, a debit or credit card, or a mobile phone to make a payment from an account—or report using the internet to pay bills or to buy something online or in a store—in the past year. Source: World Bank. Inclusive Digital Financial Services: Nepal. <https://digitalfinance.worldbank.org/country/nepal> (accessed 10 November 2025).

²³ Y. Lohia. 2024. IFAD, IOM Celebrate Vital Role of Remittances in Nepal's Development. *International Fund for Agricultural Development*. 11 June. <https://www.ifad.org/en/w/news/ifad-iom-celebrate-vital-role-of-remittances-in-nepal-development>.

²⁴ IME Pay merged with Khalti in 2025, and the combined entity operates under the name IME Khalti Limited, or the consumer-facing brand, Khalti By IME. Source: H. Agrawal. 2025. Together as One: Khalti by IME is Here. *Khalti*. 17 July. <https://blog.khalti.com/home/khaltibyime/>.

²⁵ P. Poudel. 2025. *A Study on Digital Lending and Its Impact on Access to Credit in Nepal: Opportunities, Challenges, and Pathways Forward*. https://www.researchgate.net/publication/389782358_A_Study_on_Digital_Lending_and_Its_Impact_on_Access_to_Credit_in_Nepal_Opportunities_Challenges_and_Pathways_Foward.

²⁶ Interview with NRB officials in 2025.

Limited access to finance is a challenge for micro, small, and medium-sized enterprises (MSMEs) in Nepal. Collateral requirements averaging 130% of loan value²⁷ make it difficult for MSMEs to qualify for credit. Moreover, with private credit bureau coverage at just 7% of the population,²⁸ financial institutions have access to limited data to assess creditworthiness. The regulatory framework further tightens private external borrowing and mandates procedural requirements under the Foreign Investment and Technology Transfer Act 2019, which affects the availability of funding and investment opportunities.²⁹ In 2023, one in four firms in Nepal cited access to finance as a priority area for support.³⁰ Only 16% of MSMEs secured initial funding through formal financial institutions, with most relying on informal lenders or personal savings (footnote 20).

Since the issuance of the Digital Lending Guideline 2022,³¹ which allows licensed banks and financial institutions to lend via digital platforms, including through PSPs acting as their agents, digital lending in Nepal has gained traction. Banks and fintech companies have been using transaction data from Quick Response (QR) codes and other digital channels to offer collateral-free digital loan products to merchants and to migrants traveling to foreign employment.³² With individuals allowed to borrow up to NRs500,000 (\$4,200), fintechs in the country like eSewa, F1Soft, and Khalti have disbursed NRs2.5 billion (\$18.7 million), NRs3.2 billion (\$24 million), and NRs1.8 billion (\$13.4 million), respectively (footnote 25). However, weak credit risk management, cybersecurity, and fraud controls, combined with low digital and financial literacy, have hindered access to digital lending.³³

To promote financial literacy, NRB launched a nationwide campaign in 2022 under its Financial Literacy Framework. The accompanying national survey found that overall financial literacy in Nepal stands at 57.9%. However, only 27.5% of adults achieved the minimum target score across all components—financial knowledge, financial behavior, and financial attitude. The survey results also showed wide disparities between rural and urban populations, with financial literacy rates of 23.5% and 43.1%, respectively (footnote 32), highlighting the need for targeted interventions in underserved areas. Financial literacy plays an important role in the adoption of digital payments, with a study confirming that higher financial literacy is strongly correlated with greater acceptance and use of digital payment systems in Nepal.³⁴

²⁷ The average loan to value ratio was 0.7683 in 2022, which is equivalent to a collateral requirement of about 130%. This refers to the typical amount of collateral that borrowers must provide to secure a loan, expressed as a percentage of the loan value. Source: NRB. 2024. *Financial Stability Report: Fiscal Year 2022/2023*. https://www.nrb.org.np/contents/uploads/2024/06/FSR-2022_23.pdf.

²⁸ This refers to the percentage of adults whose information is listed in private credit bureaus (nonbank lenders). Source: Statista. Private credit bureau coverage in Nepal 2015–2020. <https://www.statista.com/statistics/732429/nepal-private-credit-bureau-coverage/>.

²⁹ Government of Nepal, Ministry of Industry, Commerce, and Supplies. 2024. Foreign investment in new industries. <https://doind.gov.np/foreign-investment>.

³⁰ World Bank. 2023. Enterprise Surveys – Nepal. <https://www.enterprisesurveys.org/en/data/exploretopics/finance> (accessed 10 November 2025).

³¹ P. Khadgi. 2022. Banks in Nepal Can Now Offer Digital Loans of up to Rs. 5 Lakhs. *TechLekh*. 20 February. <https://techlekh.com/digital-lending-guidelines-nepal/>.

³² International Finance Corporation. 2025. *Digital Financial Services in Nepal*. <https://www.ifc.org/content/dam/ifc/doc/2025/digital-financial-services-in-nepal.pdf>.

³³ Interview with NRB representatives in 2024; *Business Age*. 2025. Digital Lending Gains Popularity, but with Rising Risks. 6 May. <https://newbusinessage.com/news/43660/digital-lending-gains-popularity-but-with-rising-risks/>.

³⁴ B. K. Thapa. 2025. Financial Literacy and Digital Payment System in Nepal. *Nepalese Journal of Management Research*. 5, pp. 64–72. <https://www.nepjol.info/index.php/njmgtr/article/view/75873/58231>.

The Payments Landscape

NRB developed key policies and frameworks that provided the foundation for digital payments. In 2014, NRB formulated the National Payments System Development Strategy as a comprehensive road map to modernize the country's payment system,³⁵ and established the Payment Systems Department to oversee its development. In 2018, NRB published the Payment System Oversight Framework, which emphasizes safety, operational reliability, and efficiency for prompt processing of payment transactions.³⁶ In 2019, NRB released the Retail Payments Strategy to create an enabling framework to develop digital payment services.³⁷

The Payment and Settlement Act 2019 and its accompanying Bylaw 2020 (amended in 2023) provide the legal framework for the development and regulation of payment, settlement, and clearing systems in the country. The Payment and Settlement Act 2019 outlines licensing requirements for entities engaged in payments. These entities are categorized as either PSOs or PSPs. PSOs facilitate the operation and management of payment systems, networks, and infrastructure, and ensure seamless and secure digital transactions, while PSPs facilitate digital payments, remittances, and fund transfers through various channels, which include digital wallets and payment gateways. Additionally, all of Nepal's commercial banks, development banks, finance companies, and microfinance institutions have licenses to operate as PSPs (footnote 32).

The government has incentivized adoption of digital payments through policies that require companies to pay their employees through the banking system, the disbursement of social benefits and other government payments through digital means,³⁸ and the option to pay taxes and revenues online.³⁹ In 2025, nearly 90% of government-to-person payments were digital, and about 30% of government revenues flowed through digitally enabled channels (footnote 32). The government's push toward formal, digitized solutions for remittance transfers has also enhanced digital payment adoption (Box 1).

Nepal's digital payments ecosystem is now expansive, covering everything from high-value settlement to retail transactions. According to NRB, transaction volumes are substantial, especially across mobile banking, wallets, and QR payments, supporting real-time transfers, merchant payments, e-commerce, and remittances (footnote 17).

³⁵ NRB. 2014. *National Payments System Development Strategy*. https://www.nrb.org.np/contents/uploads/2019/12/Nepal_National_Payment_System_Development_Strategy.pdf.

³⁶ NRB. 2018. *Payment System Oversight Framework*. https://www.nrb.org.np/contents/uploads/2019/12/Payment_System_Oversight_Framework_2018_20741108.pdf.

³⁷ NRB. 2019. *Retail Payments Strategy*. <https://www.nrb.org.np/contents/uploads/2019/12/Retail-Payment-Strategy-2019.pdf>.

³⁸ *Nepal Republic Media*. 2025. Finance Ministry mandates electronic payments for all public institutions from new FY. <https://www.myrepublica.nagariknetwork.com/news/finance-ministry-mandates-electronic-payments-for-all-public-institutions-f-23-86.html>.

³⁹ ConnectIPS. Government of Nepal – Driving adoption for electronic payments in Nepal, also enables NEPALPAY QR. <https://connectips.com/index.php/0429/26-govt-pay> (accessed 10 November 2025).

Box 1: Formalizing Remittances Through Digital Wallets



Remittances are a cornerstone of Nepal's economy. Yet, they have traditionally relied on informal or cash-based transfer channels that are costly, risky, and offer limited transparency. To address these inefficiencies, Nepal Rastra Bank (NRB) has prioritized the formalization and digitization of remittance transfers.

An interview with senior NRB officials highlights the success of licensed payment service providers in channeling inbound remittances directly into mobile wallets. This approach reduces transaction costs, enables instant wallet crediting, and improves accessibility for recipients, including those in remote areas. To further encourage the use of formal systems, NRB encourages banks to offer an additional 1% interest rate on deposits originating from remittances, boosting incentives to remain within regulated channels.

Another challenge lies in communication and trust-building. Many migrant workers continue to rely on informal money transfer agents offering higher exchange rates, despite the greater security and reliability of regulated options. In response, NRB has expanded digital and financial literacy and awareness campaigns to promote safer remittance practices.

Taken together, these initiatives demonstrate how the formalization of remittances widens the user base for digital payments, fosters financial inclusion, and lays the foundation for new products and services that support Nepal's ongoing digital transformation. Today, 93% of remittances are sent through formal channels, and the cost of sending \$200 as a remittance to Nepal now averages 3.7%, down from 4.7% in 2018.^a

^a *Business Age*. 2025. Remittance Realities. 9 August. <https://www.newbusinessage.com/news/44995/remittance-realities/>.

Source: Interview with NRB officials in March 2025.

The Payment Processing Infrastructure

The National Payment Switch in Nepal is the central switching infrastructure for payments in Nepal, including electronic fund transfers, card payments, QR code-based payments, and e-money. Based on open application programming interfaces (APIs) it enables interoperability between different payment systems.

The National Payment Switch came into operation in 2021 through the establishment of the Retail Payment Switch.⁴⁰ The National Clearing House Limited (NCHL), the largest PSO, manages the National Payment Switch along with other payment systems, ensuring their security and reliability (Box 2). As of 2025, 48 banks, PSOs, and PSPs were enrolled in the National Payment Switch.⁴¹ Total retail transactions cleared through the switch increased rapidly from 19.0 million in 2021 to 39.6 million in 2022, and reached 50.6 million in 2023.⁴² In particular, the ease of use of QR-based transactions has resulted in rapid adoption among retail outlets and small businesses with 2.34 million merchants using QR codes by January 2025, up from 280,000 in 2021.⁴³

⁴⁰ NCHL. Retail Payment Switch (RPS). <https://nchl.com.np/retail-payment-switch-rps/> (accessed 10 November 2025).

⁴¹ NCHL. Our Members – RPS. <https://nchl.com.np/members/> (accessed 10 November 2025).

⁴² NCHL. *Preparing Nepal's Digital Payments for Global Acceptance: Annual Report 2022–2023*. https://nchl.com.np/wp-content/uploads/2024/04/NCHL_Annual-report-79-80.pdf.

⁴³ *The Kathmandu Post*. 2025. QR Code Drives Nepal's Digital Payment Boom. 11 April. <https://kathmandupost.com/money/2025/04/11/qr-code-drives-nepal-s-digital-payment-boom>.

Box 2: The Nepal Clearing House Limited's Approach to Payment System Interoperability



Nepal's payment landscape has seen major transformations through the implementation of the National Payment Switch, led by the Nepal Clearing House Limited (NCHL). Operating as a public-private partnership between the central bank and financial institutions, NCHL has developed centralized infrastructure to address fragmentation challenges across Nepal's diverse finance sector.

The National Payment Switch consists of two core components designed to overcome technical integration challenges. The Retail Payment Switch, launched in 2021, enables interoperability for non-card payments, including Quick Response transactions. Its National Payments Interface functions as an open application programming interface stack that serves as both a standardized integration layer and a message converter, addressing the challenge of disparate data formats and messaging protocols across different banking systems.

The second component, the National Card Switch, facilitates the routing, switching, and settlement of domestic card transactions, and underpins Nepal's domestic card scheme, NEPALPAY. Through partnerships with international network providers, NEPALPAY cards are accepted globally.

This unified approach has yielded impressive results. The Retail Payment Switch now supports 100% participation from banks and financial institutions, and 77% participation from payment system operators and payment service providers. Annual transaction values processed through this system have already surpassed Nepal's gross domestic product, demonstrating substantial market adoption.

Building on this foundation, NCHL is promoting cross-border interoperability with the National Payments Interface Exchange, a cross-border switch that enables international payment connections to India and other countries. By providing standardized components like NPI Banks (on-premise integration interfaces) and message conversion capabilities based on ISO20022 standards, NCHL has created an ecosystem where players can connect through a single integration rather than developing costly bilateral connections with multiple institutions.

Source: Interview with a senior NCHL representative in December 2024.

Despite the implementation of the National Payment Switch, interoperability across payment systems is still evolving. The PSOs have established and implemented their own payment infrastructure and scheme. NCHL operates the NEPALPAY scheme with services such as NEPALPAY QR, NEPALPAY Instant, and NEPALPAY card. Similarly, other PSOs have their own scheme of services that they offer to their participating members—Fonepay Payment Service Limited operates the Fonepay scheme, Nepal Payment Solution Limited operates the Instapay scheme, and SmartChoice Technologies Limited operates the Smartpay scheme. Currently, systems of NCHL and SmartChoice Technologies are interoperable through the National Payment Switch, and work is underway to expand interoperability across other systems.

Reliability and stability of middleware systems that connect third-party payment gateways also continue to be strengthened. Middleware performance can influence transaction speed and consistency, especially for use cases such as cross-border payments that support trade.⁴⁴ As the ecosystem grows, stakeholders have noted ongoing efforts to enhance the reliability and reduce the cost of mobile banking transactions, as well as simplify processes such as wallet-to-wallet payments, which currently route through banks.⁴⁵

⁴⁴ Khatapana. 2025. How NRB's New Directive Impacts Khalti, Mobile Banking, and Digital Nepal. 18 March. <https://www.khatapana.com/blogs/442/how-nrbs-new-directive-impacts-khalti-mobile-banki>; NRB. 2024. *Payment Systems Oversight Report FY2023/24*. <https://www.nrb.org.np/contents/uploads/2025/01/Payment-Oversight-Report-2023-24.pdf>.

⁴⁵ United Nations Capital Development Fund. 2022. *Report on Inclusive Innovation Strategies in Migrant Remittances and Financial Services: Nepal Country Assessment*. <https://migrantmoney.uncdf.org/resources/research/nepal-country-assessment-report-on-inclusive-innovation-strategies-in-migrant-remittances-and-financial-services/>.

These developments reflect a broader transition from separate systems toward a more integrated architecture. According to a representative from the Nepal Digital Payments Company, payment aggregators in Nepal do not yet function as end-to-end solutions that can seamlessly connect or process transactions through a single unified interface. As a result, PSPs continue to integrate individually with multiple systems (Box 3).⁴⁶ This evolving architecture shapes the efficiency of cross-border transactions, an area where continued progress will further enhance the ease and reliability of international digital payments.

Box 3: Technical Integration Challenges in Nepal's Digital Payment Ecosystem



The absence of a strong payment aggregator poses a barrier to payment system interoperability in Nepal, according to representatives of the Nepal Digital Payments Company. As a senior representative at the company explains:

“There are around four core payment systems that banks and other financial institutions use...integrating with each of them is not possible.”

Integrating separately with each payment system operator increases security risks and inefficiencies in time and cost. This fragmentation also creates multiple technical challenges:

- Different system architectures that require different integration approaches.
- Application programming interfaces (APIs) that are not standardized or aligned with international norms.
- Absence of a centralized testing and development environment.

The aggregator model, which can potentially streamline payment processes, is not yet functioning as intended. Unlike other markets where aggregators provide unified access, in Nepal, “even if we integrate with aggregators, we still have to connect individually with each API under that aggregation,” notes the representative.

These technical barriers are compounded by cost pressures:

- High security compliance requirements despite limited revenue.
- Need for advanced fraud detection systems.
- Required investment in artificial intelligence and machine learning capabilities.
- Limited transaction volumes due to market size.

“The investment required for the technical infrastructure is a very big cost for us,” adds the representative. “Even though we have technical readiness, implementation is not possible because we do not have the financial readiness.”

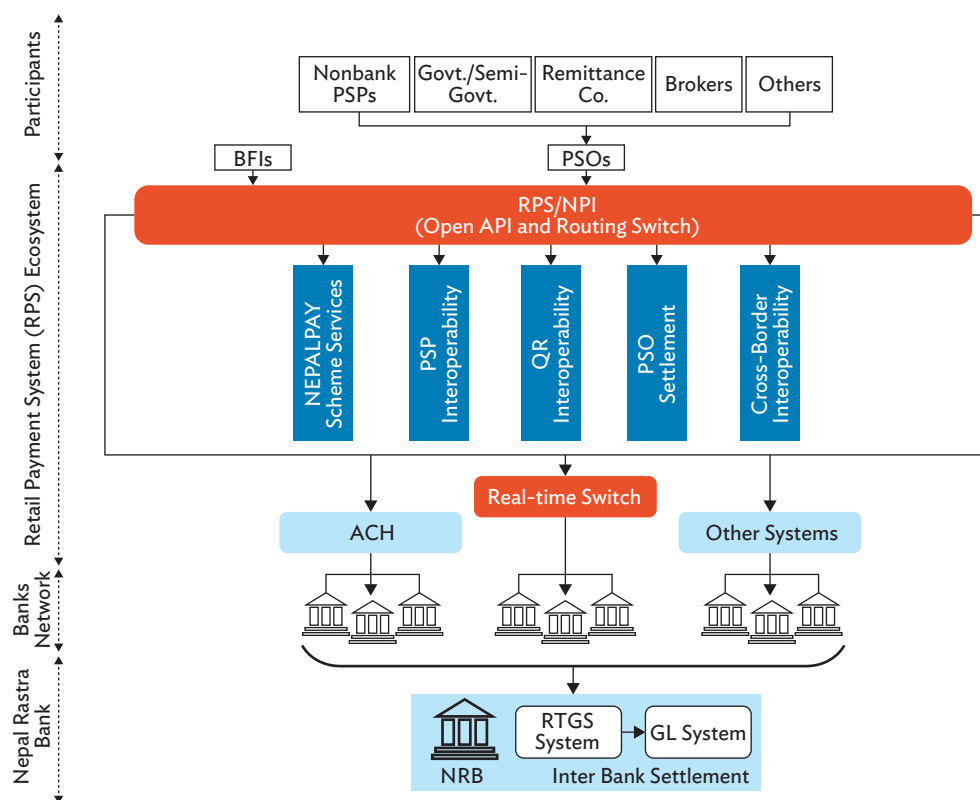
These insights suggest that addressing Nepal's payment integration challenges will require not only technical standards, but also new approaches to sharing infrastructure costs across the ecosystem.

Source: Interview with Nepal Digital Payments Company representatives in December 2024.

⁴⁶ Interview with Nepal Digital Payments Company representatives in 2024.

To address fragmentation across payment systems, NRB published the National Payment Switch and the National Payment Ecosystem Master Reference Document that prioritizes system-wide interoperability, both domestically and internationally (Figure 1).⁴⁷ QR interoperability is guided by the NepalQR Standardization Framework and Guidelines.⁴⁸ However, while technical progress on interoperability is advancing, building a common understanding among stakeholders on what interoperability means in practice remains equally important (Box 4).

Figure 1: High-Level Architecture of the Retail Payment Switch



ACH = automated clearing house, API = application programming interface, BFIs = banks and financial institutions, Co. = company, GL = general ledger, Govt = government, NPI = National Payments Interface, NRB = Nepal Rastra Bank, PSOs = payment system operators, PSP = payment service providers, QR = Quick Response, RTGS = real time gross settlement.

Source: NRB. 2025. *National Payment Switch (NPS) and the National Payment Ecosystem Master Reference Document*. October. <https://www.nrb.org.np/contents/uploads/2025/10/National-Payment-Switch-NPS-and-the-National-Payment-Ecosystem-Master-Reference-Documents-2025.pdf>.

⁴⁷ NRB. 2025. *National Payment Switch (NPS) and the National Payment Ecosystem Master Reference Document*. October. <https://www.nrb.org.np/contents/uploads/2025/10/National-Payment-Switch-NPS-and-the-National-Payment-Ecosystem-Master-Reference-Documents-2025.pdf>.

⁴⁸ NRB. 2020. *NepalQR Standardization Framework and Guidelines*. <https://www.nrb.org.np/contents/uploads/2021/01/QR-Code-Guidelines-and-Framework-and-Specifications.pdf>.

Box 4: Diverging Perspectives on Payment System Interoperability in Nepal



Discussions on payment system interoperability in Nepal show that stakeholders hold varying views on its effectiveness. While some multilateral agency assessments and market participants point to interoperability challenges, Fonepay—one of Nepal’s largest payment system operators—strongly contests this view.

“We are fully interoperable—all commercial banks operating in Nepal can interoperate through our network,” explains Fonepay’s chief executive officer. “Right now, any bank can send money to another bank through our infrastructure. We are an open-loop network, not a closed-loop system.”

This divergence between external assessments and operator perspectives highlights several important considerations:

- The need for clear, agreed definitions of interoperability in policy discussions.
- Potential gaps between technical capabilities and practical implementation.
- The importance of distinguishing between different types of interoperability (bank-to-bank, wallet-to-wallet, cross-border).
- How varying interpretations of interoperability can affect policy recommendations.

Understanding these different perspectives is crucial for policymakers working to enhance payment system integration. Future initiatives should therefore focus not only on expanding technical capabilities, but also on promoting a common understanding and standardized measurement of interoperability across the ecosystem.

Source: Interview with Fonepay representatives in December 2024.

For cross-border payments, the National Payments Interface Exchange is a cross-border switch that is currently functional for person-to-merchant payments, while peer-to-peer payments are being tested (footnote 47).

In 2024, NRB and the Reserve Bank of India signed a regulatory term of reference to facilitate the integration of Nepal’s National Payments Interface with India’s Unified Payments Interface (UPI).⁴⁹ Since March 2024, Indian travelers can use their own payment systems to pay merchants in Nepal who have Fonepay or Khalti QR codes. This implementation has shown promising growth, with daily transactions increasing from approximately 500 at launch to 2,000 by early 2025, representing a total daily transaction value of NRs6 million (\$42,000) and cumulative transactions worth NRs1.6 billion (\$11 million) since launch.⁵⁰ As of January 2025, NEPALPAY QR became accessible to tourists from the People’s Republic of China, Italy, the Republic of Korea, Malaysia, and Singapore.⁵¹

While these represent achievements, the reciprocal functionality enabling Nepalis to make QR payments in India and other countries remains pending, highlighting continued challenges in achieving fully bidirectional payment integration. While Indians can make QR payments in Nepal (with a standard charge of 1.95% per transaction), the reciprocal service for Nepalis in India faces delays primarily due to unresolved issues around commission structures (footnote 50). In India, no charges are levied on QR code payments, creating uncertainty regarding

⁴⁹ Reserve Bank of India. 2024. Press Release: Terms of Reference signed between Reserve Bank of India and Nepal Rastra Bank on Integration of Unified Payments Interface (UPI) of India with National Payments Interface (NPI) of Nepal for cross-border remittances. 15 February. https://www.rbi.org.in/Scripts/BS_PressReleaseDisplay.aspx?prid=57332.

⁵⁰ The Kathmandu Post. 2025. Nepalis still wait for QR payments in India a year after India’s launch in Nepal. 21 March. <https://kathmandupost.com/money/2025/03/21/nepalis-still-wait-for-qr-payments-in-india-a-year-after-india-s-launch-in-nepal>.

⁵¹ Nepal Republic Media. 2025. NCHL unlocks seamless QR payments for tourists from 10 countries. 30 January. <https://myrepublica.nagariknetwork.com/news/nchl-accelerates-cross-border-digital-payments-26-92.html>.

who would bear the cost-of-service commissions for Nepali banks when their customers make payments in India. This situation highlights how differences in fee structures and regulatory approaches between countries can impede the implementation of otherwise technically feasible cross-border payment solutions.

Although QR payments and mobile banking have grown rapidly in Nepal, the broader payment ecosystem is still transitioning from legacy routing arrangements toward more streamlined pathways. Nepal's current dependence on traditional banking channels for international transactions—together with existing regulatory frameworks governing cross-border capital flows—shapes the cost and operational steps involved in facilitating cross-border payments and remittances (footnote 45). As digital payment systems continue to mature, there is significant potential to simplify these processes and enhance the overall efficiency of international transactions. With trade and remittance flows between Nepal and its neighbors continuing to expand, Nepal–India financial exchanges represent an untapped opportunity for advancing digital payments (Box 5).

Box 5: The India–Nepal Payment Corridor Opportunity

The economic relationship between India and Nepal creates a digital payment opportunity that remains largely untapped. Each year, billions of dollars move between the two countries—mainly through trade and remittance flows—yet, most transactions continue to rely on traditional banking channels, and digital payment integration remains limited.

Indian visitors can now make payments at select Nepali merchants using QR codes compatible with India's Unified Payments Interface, but this arrangement remains small in scale and primarily supports inbound transactions. Similarly, the Nepal Clearing House Limited (NCHL) has begun exploring peer-to-peer and remittance linkages with India, but these efforts are in the early stages. Efforts to introduce more advanced solutions, such as blockchain-based trade finance systems, are still at the conceptual stage.

Strengthening the digital payment corridor between India and Nepal through interoperable standards and shared infrastructure can generate efficiency gains for businesses and individuals in both nations. Such an initiative can serve as a foundation for broader regional payment integration within the South Asia Subregional Economic Cooperation framework.

Source: Interview with Sanjib Subba, Kathmandu Fintek in December 2024.

The Digital Landscape: Connectivity, Access, and Affordability

Nepal's mountainous terrain makes infrastructure deployment both difficult and costly. Limited direct access to submarine cables that carry the bulk of global data traffic leads to high international connectivity costs and reduced network reliability, ultimately resulting in higher prices for end users.⁵²

However, a notable achievement is Nepal's investment in fiber-optic networks across the country, now covering all 77 districts.⁵³ Data indicates that 3G coverage reached 90% of the population and 4G reached 86% of the population in 2024.⁵⁴ By 2023, 56% of individuals were estimated to be using the internet in Nepal, and 75% of individuals owned a mobile phone (footnote 54). Data shows that 57% of total mobile connections are via

⁵² World Bank. 2022. *Project Appraisal Document on Digital Nepal Acceleration (DNA) Project*. <https://documents1.worldbank.org/curated/en/995101654266502751/pdf/Nepal-Digital-Nepal-Acceleration-DNA-Project.pdf>.

⁵³ *Nepali Telecom*. 2024. Nepal Telecom brings FTTH in 77 districts after Humla launch. 16 August. <https://www.nepalitelecom.com/nepal-telecom-ftth-availability>.

⁵⁴ International Telecommunication Union. Data Hub: Nepal. <https://datahub.itu.int/data/?e=NPL> (accessed 10 November 2025).

smartphones, reflecting the growing prevalence of smart devices across the country.⁵⁵ Mobile banking usage has surged dramatically, particularly after the COVID-19 pandemic, growing at a compound annual rate of 58% over 8 years (footnote 20).

To ensure inclusive digital access across the country, continued enhancements in last-mile connectivity, network service quality, device and data affordability, digital literacy, and rural uptake will be important. While Nepal's fiber-optic backbone now spans all districts, about half of the population lives more than 10 kilometers from core fiber routes,⁵⁶ which can influence the quality of service delivery. Strengthening last-mile connectivity will help maximize the benefits of the country's growing digital infrastructure.

Regarding affordability, the cost of mobile broadband decreased substantially in relation to average gross national income (GNI) per capita, to 2.05% in 2024, only slightly above the United Nations Broadband Commission's affordability target of 2% GNI per capita. Fixed broadband, on the other hand, remains less affordable at a cost of 7% of average monthly GNI per capita (footnote 54). Moreover, the affordability of entry-level smartphones has deteriorated in recent years—amounting to 33% of monthly income in 2023. A major factor contributing to the decline in affordability is the high tax burden on phones, which reached 71% in 2023, far exceeding that of Bangladesh (0%), India (3%), and Sri Lanka (30%) (footnote 56).

As a result, there are wide variations in internet usage across provinces, rural and urban areas, and among different gender, age, and income groups. For instance, Bagmati leads in internet usage rates at 60% of households while Madhesh lags at 14%. Moreover, 49.3% of urban households use the internet compared to 17.5% of rural households, limiting the potential benefits of digital trade facilitation for rural businesses and producers (footnote 56).

Similar trends are reflected in the adoption of digital payments. The share of adults making or receiving digital payments in Nepal almost tripled from 10% in 2014 to 29% in 2021. However, this is still lower than in other lower-middle income countries and South Asian peers, where digital payment adoption rates were 38% and 34% on average, respectively (footnote 56). Growth in digital payment usage has been faster in urban areas, where 34% of adults have made or received a digital payment, compared with only 27% in rural areas (footnote 20).

To address the digital divide, some PSPs have introduced offline QR payment solutions, such as FoneTag and NepalPayTag, allowing transactions where merchants are online but customers are offline. This innovation uses near-field communication technology, enabling customers to make payments by tapping their phones on a point-of-sale device without an internet connection.⁵⁷

⁵⁵ GSMA. 2023. *IoT and Essential Utility Services: Nepal market case study*. <https://www.gsma.com/solutions-and-impact/connectivity-for-good/mobile-for-development/wp-content/uploads/2023/04/IoT-and-Essential-Utility-Services-Nepal-market-case-study.pdf>.

⁵⁶ World Bank. 2025. *Unlocking Nepal's Growth Potential*. <https://documents.worldbank.org/en/publication/documents-reports/document-detail/099032125103030263>.

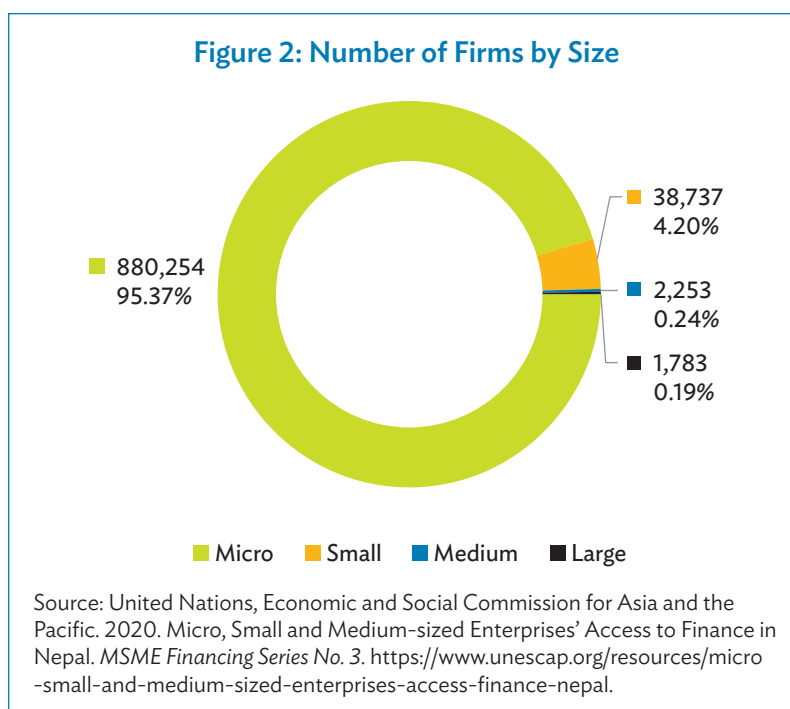
⁵⁷ Alliance for Financial Inclusion. 2024. *Payment Innovations and Risks in South Asia*. https://www.afi-global.org/wp-content/uploads/2024/10/Payment-Innovations-and-Risks-South-Asia_April.pdf.

The Digital Readiness of Businesses

Nepal's economy is characterized by a large informal sector, accounting for approximately 37.5% of economic activity.⁵⁸ MSMEs form a major part of this landscape (Figure 2), with over 920,000 establishments recorded in the 2018 National Economic Census, nearly half of which are unregistered and operate informally.⁵⁹

Among formal MSMEs in Nepal, at least 90% were reported to have high-speed fixed-broadband internet in 2023. However, only half of the firms had an online presence, and less than 10% of firms made payments electronically in 2023 due to various challenges (footnote 56).

A key challenge is the high cost associated with digital payments.⁶⁰ Digital payment transactions in Nepal typically incur transaction fees ranging from NRs2.00 to NRs8.00 (\$0.015 to \$0.056) for domestic transfers, depending on the platform and transaction amount.⁶¹ For small-value transactions, these fees can represent a significant percentage of the payment, discouraging both small merchants and consumers from using digital channels. These costs are exacerbated in cross-border transactions due to foreign exchange fees and conversion charges, posing major hurdles for businesses aiming to engage internationally.



⁵⁸ L. Medina and F. Schneider. 2018. Shadow Economies Around the World: What Did We Learn Over the Last 20 Years? *IMF Working Paper*. WP/18/17. International Monetary Fund. <https://www.elibrary.imf.org/view/journals/001/2018/017/article-A001-en.xml>.

⁵⁹ United Nations, Economic and Social Commission for Asia and the Pacific. 2020. Micro, Small and Medium-sized Enterprises' Access to Finance in Nepal. *MSME Financing Series No. 3*. <https://www.unescap.org/resources/micro-small-and-medium-sized-enterprises-access-finance-nepal>.

⁶⁰ A. Pathak. 2024. Digital Payment in Nepal: An Overview and Recommendations. *Rupandehi Campus*. 4 (1). <https://www.nepjol.info/index.php/rcj/article/view/62916>.

⁶¹ ConnectIPS. 2021. Digital Payment. <https://connectips.com/knowmore/digital-payment.php> (accessed 10 November 2025).

Many MSMEs have limited technical expertise and operational infrastructure required to integrate and manage advanced digital payment systems. Only one in five firms in Nepal reported they started or expanded their use of digital platforms during the COVID-19 pandemic.⁶² This limited capacity is particularly challenging for cross-border payments, which require more sophisticated technical knowledge, security protocols, and compliance with international standards.

Moreover, as cyberthreats become increasingly complex,⁶³ MSMEs generally have limited financial resources and technical expertise to deploy advanced cybersecurity measures, such as the use of artificial intelligence (AI) for fraud detection. These systems—featuring tools like biometric verification, behavioral analytics, and real-time data monitoring—require significant investment and expertise that are typically beyond the reach of MSMEs.⁶⁴

This gap is further compounded by Nepal's shortage of digital talent as skilled engineers and developers opt for opportunities abroad.⁶⁵ Industry stakeholders confirm the human resource constraints as skilled professionals leave the country for better opportunities, while those who stay often demand salaries beyond what local companies can afford.⁶⁶

Another barrier is the informality of many MSMEs, which limits their access to technology and formal financial and payment systems. Consequently, informal MSMEs are less likely to adopt digital payment platforms and other advanced financial technologies, which presents a barrier to their integration into the digital economy.⁶⁷

Policies and Regulations for Digital Financial Services

Payment System Licensing and Oversight Framework

The licensing of PSOs and PSPs has been pivotal in transforming the country's digital payment landscape. The licensing requirements for entities operating payment systems or providing payment services in Nepal are outlined in the Payment and Settlement Act of 2019 and related regulations. NRB has established specific criteria that must be met to qualify for these licenses, including business registration, minimum capital requirements, and procedures for managing various risks.

As of October 2025, NRB had 29 licensed payment companies—8 PSOs and 21 PSPs.⁶⁸ These licenses have been granted to a diverse range of entities, including banks, nonbank financial institutions, telecommunication providers, and fintech companies. By opening the market to both bank and nonbank entities, NRB's licensing framework has enhanced competition, diversified service offerings, and expanded the reach of digital payment systems across the country.

⁶² World Bank. 2021. COVID-19 Business Pulse Survey Dashboard. <https://www.worldbank.org/en/data/interactive/2021/01/19/covid-19-business-pulse-survey-dashboard>.

⁶³ A. Ray. 2025. Cybercrimes in Nepal have become more sophisticated and organised. *The Kathmandu Post*. 14 April. <https://kathmandupost.com/interviews/2025/04/14/cybercrimes-in-nepal-have-become-more-sophisticated-and-organised>.

⁶⁴ Interviews with fintech companies in 2024.

⁶⁵ *The Kathmandu Post*. 2022. The future of Nepal's tech industry. 21 September. <https://kathmandupost.com/columns/2022/09/21/the-future-of-nepal-s-tech-industry>.

⁶⁶ Interviews with fintech companies in 2024.

⁶⁷ NRB. 2021. *Financial stability: Challenges and opportunities in the post-Covid context*. <https://www.nrb.org.np/contents/uploads/2023/06/2021-Financial-Stability-Challenges-and-Opportunities-in-the-post-COVID-context.pdf>.

⁶⁸ NRB. 2023. *List of Payment System Operator (PSO) and Payment Service Provider (PSP) (As of 2082/06/28)*. <https://www.nrb.org.np/contents/uploads/2025/10/List-of-PSOPSP-in-Asoj-29-Eng.pdf>.

Nepal's regulatory framework for digital payments has advanced in recent years, with NRB introducing a series of measures to promote innovation in digital payments while ensuring system security and reliability, as laid out in its Strategic Plan of 2022–2026.⁶⁹ Key instruments include the Payment System Oversight Manual 2021, Payment System Inspection and Supervision Bylaws 2021, Cyber Resilience Guidelines 2023, Procedure for Onsite Inspection of Payment-related Activities of Banks and Financial Institutions 2024, and Unified Payment System Directive 2025—all aimed at strengthening oversight, safety and security, operational resilience and efficiency (footnote 32). NRB can enforce compliance through fines, suspension of services, or revocation of licenses.

The licensing process continues to be enhanced, and industry stakeholders have highlighted opportunities to streamline procedures and strengthen clarity around the respective roles of banks and PSPs to support equal treatment for smaller providers and new entrants.⁷⁰ Moreover, the current licensing framework is still adapting to emerging and innovative digital business models, presenting scope for further refinement. For instance, the concept of aggregator is not formally used in NRB's licensing taxonomy, although some PSOs may act as aggregators by integrating multiple PSPs and banks for interoperability and switching services. Likewise, although emerging products such as buy-now-pay-later and other digital lending services are referenced in NRB's oversight reports, no dedicated licensing framework existed for such fintech products as of 2025.

NRB's Payment System Oversight Framework is principle-based, aligning with the Principles for Financial Market Infrastructures developed by the Committee on Payments and Market Infrastructures and the International Organization of Securities Commissions (footnote 36). Within this framework, inspection schedules are risk-based. Licensed PSOs and PSPs require on-site inspections at least once every 2 years, with special or follow-up inspections conducted when off-site monitoring indicates emerging concerns.⁷¹ Banks are audited annually with the Bank Supervision Department assessing each bank's capital, governance, and risk controls to determine the scope and frequency of onsite inspections.⁷²

While this framework represents a modern, principles-based oversight regime, several challenges have been identified in a review by the International Monetary Fund.⁷³ Some key elements—such as provisions related to the finality and irrevocability of transactions, and formal system designation—have yet to be fully embedded in law. The capacity of the Payment Systems Department needs to be strengthened. Moreover, NRB's dual role as regulator and shareholder in core payment infrastructure has raised conflict-of-interest concerns. The biennial inspection cycle, combined with bank-level anti-money laundering, cyber resilience, and consumer protection requirements, creates high compliance costs that smaller fintechs struggle to meet, which can discourage market entry, limit innovation, and contribute to operational vulnerabilities.

⁶⁹ NRB. 2022. *Fourth Strategic Plan 2022–2026*. <https://www.nrb.org.np/contents/uploads/2022/03/SP-4-Text-for-website.pdf>.

⁷⁰ Interviews with fintech companies in 2024.

⁷¹ K. Prasain. 2021. Central bank rolls out Payment System Inspection and Supervision Bylaw. *The Kathmandu Post*. 2 September. <https://kathmandupost.com/money/2021/09/02/central-bank-rolls-out-payment-system-inspection-and-supervision-bylaw>.

⁷² NRB. 2025. *Bank Supervision Report*. <https://www.nrb.org.np/contents/uploads/2025/03/Annual-Bank-Supervision-Report-2024-4.pdf>.

⁷³ International Monetary Fund. 2023. *Nepal Technical Assistance Report – Financial Sector Stability Review*. <https://www.imf.org/en/Publications/CR/Issues/2023/10/03/Nepal-Technical-Assistance-Report-Financial-Sector-Stability-Review-539970>.

Cybersecurity, Personal Data Protection, and Consumer Protection

Cybersecurity

NRB has introduced measures to enhance cybersecurity in the finance sector, including the issuance of the Cyber Resilience Guidelines in 2023. The guidelines require all licensed entities operating in the payment ecosystem to enhance their ability to identify, assess, and mitigate cyber risks associated with digital transactions. Key actions include conducting regular risk assessments, developing incident response plans, implementing security controls and encryption, and regular employee training on cybersecurity.⁷⁴

As Nepal's cybersecurity framework continues to mature, several areas for improvement have been identified. For instance, establishing a mechanism for periodic review and updates of the guidelines will help ensure continued alignment with regional and international standards and practices, such as payment tokenization for stored card credentials.⁷⁵

In addition, licensed entities are at different stages of readiness to implement the guidelines, reflecting variations in technical capacity and operational resources.⁷⁶ Smaller PSPs, in particular, have highlighted the value of tailored support, capacity-building programs, and risk-based compliance approaches that can enable them to meet cybersecurity requirements while continuing to innovate and grow.⁷⁷

Data Protection

Nepal's Privacy Act 2018 governs the protection of personal data and privacy. Analysis observes some areas for improvement to fully align with global best practices like the European Union's General Data Protection Regulation. Key global-standard principles, such as explicit data subject rights (e.g., the right to data portability and the right to be forgotten), stringent rules for cross-border data transfers, and the establishment of an independent data protection authority are not as comprehensively defined in Nepal's framework.⁷⁸ Other identified areas for improvement related to personal data protection and privacy include the following (footnote 52):

- **Privacy by design:** There is a need to embed privacy considerations into the development and design of digital platforms from the outset, ensuring that personal data protection is an integral part of system functionality.
- **Data storage limitations:** Clear guidelines on the duration for which personal data can be stored need to be developed. Establishing defined limits is crucial to prevent unnecessary retention and reduce the risk of data breaches or misuse.
- **Algorithmic decision-making oversight:** The Privacy Act does not yet include explicit regulatory provisions on automated decision-making processes. As Nepal advances its use of digital technologies, including AI, there is an opportunity to further strengthen the privacy and data governance framework. Enhancing transparency and accountability in automated decision-making processes is vital to protect against bias and ensure fairness.

⁷⁴ NRB. 2023. *Cyber Resilience Guidelines 2023*. <https://www.nrb.org.np/contents/uploads/2023/08/Cyber-Resilience-Guidelines-2023.pdf>.

⁷⁵ NRB statutes, bylaws, unified directives, and oversight reports that govern retail payments show no clause that makes card-on-file tokenization compulsory; Deloitte. 2023. *Network Tokenization for Merchants*. <https://www.deloitte.com/content/dam/assets-zone3/us/en/docs/industries/financial-services/2024/us-fsi-deloitte-network-tokenization-for-merchants.pdf>.

⁷⁶ *Nepal Republic Media*. 2024. Majority of BFIs struggle to enforce cyber safety measures on the digital transactions: Experts. 19 February. <https://myrepublica.nagariknetwork.com/news/majority-of-bfis-struggle-to-enforce-cyber-safety-measures-on-the-digital-transactions-experts>.

⁷⁷ Interview with fintech companies in 2024.

⁷⁸ RA & Associates. Data Protection Law in Nepal. <https://www.raandassociate.com/data-protection-law-in-nepal/> (accessed 10 November 2025).

These areas create uncertainties for businesses and users, slowing the growth of trust in the digital economy and impacting cross-border trade.

Consumer Protection

NRB has established a financial consumer protection framework, including the Financial Consumer Protection and Grievance Management Manual in 2020. This framework sets out provisions to safeguard consumer rights, promote fairness and transparency, and ensure accountability in financial service delivery. It also outlines mechanisms for grievance handling and conflict resolution.⁷⁹ To support implementation, NRB operates a dedicated online grievance management system to receive and address public inquiries, complaints, and suggestions related to financial services.

However, the consumer protection framework for digital payments, particularly for cross-border transactions, is still evolving. A more comprehensive approach is needed to define clear dispute resolution mechanisms for cross-border payments, establish customer liability limits to protect consumers from unauthorized international transactions, and strengthen oversight to ensure consistent protection across both domestic and cross-border digital payment channels.

Innovation and Emerging Technologies

Digital Identity and Know-Your-Customer Framework

Nepal is committed to improving and integrating its national identification system into various services, including payments, as part of its digital public infrastructure (DPI) (footnote 12). Currently, the country's digital national identification system is still at a nascent stage, which creates challenges for seamless digital onboarding and know-your-customer (KYC) verification, particularly in cross-border transactions (footnote 32). In 2022, the telecommunication regulator, Nepal Telecommunications Authority, in coordination with NRB, deployed a centralized, biometric-based KYC system for the registration of Subscriber Identity Module (SIM) cards leveraging the national identification database.⁸⁰ This initiative is viewed as a first step toward building a national digital identification infrastructure (footnote 32).

In early 2025, the government launched the E-Governance Blueprint and integrated the national identification system into the Nagarik App, its flagship platform for digital public services.⁸¹ From this point, use of national identification became mandatory for key services such as banking and passport issuance, and pilots are under way to plug the national identification APIs directly into finance sector KYC workflows.⁸² These initiatives mark the formation of a DPI stack in Nepal, enabling centralized identity verification and data interoperability across sectors.

⁷⁹ Alliance for Financial Inclusion. 2022. Member Series: Nepal's Financial Inclusion Journey Towards Access to Finance, Financial Literacy and Financial Consumer Protection. <https://www.afi-global.org/publication/member-series-nepals-financial-inclusion-journey-towards-access-to-finance-financial-literacy-and-financial-consumer-protection/>.

⁸⁰ Nepal Telecommunications Authority. 2022. *Expression of Interest: Design, Development/Supply, Integration, Implementation, Operation and Support for Centralized Biometric based Digital Know Your Customer (DigitalKYC)*. <https://nta.gov.np/uploads/contents/EOI-KYC.pdf>.

⁸¹ A. Macdonald. 2025. Nepal integrates national ID with digital govt app to expand public service access. *Biometric Update*. 30 January. <https://www.biometricupdate.com/202501/nepal-integrates-national-id-with-digital-govt-app-to-expand-public-service-access>.

⁸² *The Kathmandu Post*. 2025. Supreme Court clears way for National ID card implementation, but people continue to face hassles. 2 February. <https://kathmandupost.com/national/2025/02/02/supreme-court-clears-way-for-national-id-card-implementation-but-people-continue-to-face-hassles>.

To balance broad access with regulatory compliance, NRB has implemented a tiered KYC approach.⁸³ Under this system, transactions below specific thresholds benefit from simplified identity checks, thereby streamlining onboarding and the usage of digital channels. Conversely, higher-value or higher-risk transactions remain subject to enhanced KYC protocols. NRB officials described this tiered model as a vital mechanism for lowering entry barriers for underserved segments while mitigating risks.⁸⁴

Against this backdrop, the potential of electronic KYC for international transactions will depend on further regulatory reforms, such as legal recognition of electronic KYC and electronic signatures, and the development of backend systems for identity verification.⁸⁵

Central Bank Digital Currency

NRB has been exploring the potential implementation of a Central Bank Digital Currency (CBDC) in Nepal since 2022. NRB has released a concept report⁸⁶ and a prototype,⁸⁷ with the aim to accomplish a CBDC pilot by 2026 (footnote 69). NRB is moving forward cautiously with its design and rollout. Its priority areas include closing regulatory and payment system gaps, strengthening cybersecurity, raising digital and financial literacy levels, and ensuring seamless interoperability with existing payment infrastructure (footnote 87). The CBDC will impact the future of cross-border payments by potentially reducing dependence on correspondent banking relationships and simplifying currency conversion processes in international trade.

Regulatory Sandbox

NRB has laid the foundation for a regulatory sandbox by launching a Digital Finance Innovation Hub in March 2025 to boost fintech innovation in Nepal, especially in the payments and digital finance space. The Hub is positioned as a structured regulatory environment for testing new digital finance products, enabling fintech companies to propose solutions to NRB and receive regulatory guidance.⁸⁸

Financial service providers and fintech companies are driving innovation within the existing regulatory framework. For instance, Fonepay has launched a digital-first credit product—a virtual credit card issued via mobile banking apps—that is seamlessly integrated with its payment network, enhancing user convenience and expanding access to credit.⁸⁹ However, industry stakeholders highlight that a more structured approach, such as a dedicated regulatory sandbox, is essential to accelerate innovation and attract investment, as well as offer regulators critical insights to inform future policies.⁹⁰

⁸³ NRB. 2019. *Retail Payments Strategy, 2019*. <https://www.nrb.org.np/contents/uploads/2019/12/Retail-Payment-Strategy-2019.pdf>.

⁸⁴ Interview with NRB officials in 2024.

⁸⁵ Interview with NRB officials in 2024.

⁸⁶ NRB. 2022. *Central Bank Digital Currency (CBDC): Identifying appropriate policy goals and design for Nepal – A Concept Report*. <https://www.nrb.org.np/contents/uploads/2022/10/CBDC-for-Nepal.pdf>.

⁸⁷ *Fiscal Nepal*. 2024. Nepal Rastra Bank's prototype for digital currency 'CBDC' signals shift towards a cashless economy. 9 August. <https://www.fiscalnepal.com/2024/08/09/17536/17536/>.

⁸⁸ K. Prasain. 2025. Nepal Rastra Bank launches digital finance innovation hub to boost fintech. *The Kathmandu Post*. 28 March. <https://kathmandupost.com/money/2025/03/28/nepal-rastra-bank-launches-digital-finance-innovation-hub-to-boost-fintech>.

⁸⁹ Fonepay. 2025. Fonepay credit card. <https://fonepay.com/customers/fonepay-credit-card>.

⁹⁰ Interview with Fonepay representatives in 2024.

Digital Solutions for Trade

Nepal has been working to streamline trade processes through the adoption of digital solutions, notably the development of the Nepal National Single Window. Launched in 2021, it provides a single-entry online platform for importers and exporters to submit required trade documents and access regulatory services across multiple agencies.⁹¹ Another milestone is the implementation of the ASYCUDA World system for digitized customs clearance.⁹² These initiatives pave the way for the future integration of digital payment systems, which can enable faster and more secure paperless trade transactions.

Regulations for Cross-Border Trade

Foreign Exchange Regulations

NRB maintains strict foreign exchange controls, with limited convertibility of the Nepalese rupee. For cross-border transactions, financial service providers typically use correspondent banking relationships with foreign banks to settle transactions. The Foreign Exchange (Regulation) Act 2019 and the Nepal Rastra Bank Act 2002 govern these transactions, imposing various requirements and restrictions that can affect the efficiency of cross-border digital payments.⁹³

NRB maintains a foreign exchange regime with restrictions primarily focused on the capital account side, while current account transactions remain largely open. Certain foreign exchange transactions continue to require prior approval and are subject to regulatory conditions, including the movement of local currency across borders. While these measures are intended to support prudent macroeconomic and reserve management, they may also create procedural requirements and processing delays for some businesses engaging in cross-border transactions, particularly MSMEs.

AML/CFT Requirements

Nepal has made progress in strengthening its anti-money laundering (AML) and combating the financing of terrorism (CFT) framework, which is essential for maintaining correspondent banking relationships and ensuring Nepali businesses can access international payment networks necessary for cross-border trade. The country has enacted the Asset (Money) Laundering Prevention Act 2008 and established the Financial Intelligence Unit within NRB to monitor transactions. The AML/CFT framework mandates comprehensive KYC and customer due diligence procedures, transaction record-keeping, and reporting from financial service providers, particularly for suspicious or unusual cross-border wire transfers.

⁹¹ P. Sharma and K. Dahal. 2023. Nepal National Single Window: Towards a Paperless Trade Regime. *UNNExT Working Paper Series*. No. 4. United Nations Economic and Social Commission for Asia and the Pacific. <https://www.unescap.org/kp/2023/nepal-national-single-window-towards-paperless-trade-regime>.

⁹² ASYCUDA World is an advanced computerized customs management system developed by the United Nations Conference on Trade and Development that streamlines and automates international trade procedures. It has become the global standard for customs automation due to its comprehensive coverage; compliance with international standards; customizability to national requirements; and ability to facilitate trade efficiency, improve revenue collection, enhance security, and promote transparency in customs operation.

⁹³ NRB. 2002. *Nepal Rastra Bank Act, 2058*. <https://www.nrb.org.np/lgd/nepal-rastra-bank-act-2002-consolidated-version-in-english>; Imperial Law Associates. 2019. *Overview of Foreign Exchange Law in Nepal (2081)*. <https://www.lawimperial.com/foreign-exchange-law-in-nepal/>.

NRB has taken steps to strengthen AML/CFT measures to align with the recommendations of the Financial Action Task Force (FATF), the global standard-setting body for combating money laundering and terrorism financing. Measures include conducting AML/CFT institutional risk profiles for all commercial banks and on-site inspections of banks identified as high risk.⁹⁴ An amendment to the Asset (Money) Laundering Prevention Act, enacted in 2024, mandates entities with share capital to declare their ultimate beneficial owners to the Registrar of Companies (footnote 95). This measure aims to enhance transparency and regulatory compliance.

Despite these advancements, implementation challenges persist, particularly in digital transaction monitoring due to limited institutional capacity, inadequate infrastructure, low awareness, and weak enforcement among smaller institutions.⁹⁵ Strengthening identity verification systems remains a critical priority for AML/CFT compliance (footnote 96). At the same time, it is essential to ensure that AML/CFT measures are proportionate and do not impede legitimate payment flows. Striking this balance is vital for trade facilitation, as overly stringent reporting requirements can create barriers to cross-border transactions, particularly for MSMEs with limited compliance resources.

Following the 2023 mutual evaluation of Nepal's AML/CFT framework, the country was placed under FATF observation. In February 2025, Nepal was added to the FATF gray list based on the need for additional progress in implementing the recommended actions. Over the next 2 years, demonstrating steady improvements in the implementation of relevant legal, policy, and institutional reforms will be important to prevent further downgrade to the FATF blacklist, which can have severe implications for international transactions and trade.⁹⁶

Key Challenges

While Nepal has made significant strides in developing its digital payment ecosystem, several critical challenges hinder further development and broader adoption of digital payments. These challenges are summarized in Table 1.

Together, these challenges highlight the need for a coordinated approach to developing Nepal's digital payment ecosystem. Addressing these challenges requires balanced attention to infrastructure development, regulatory reform, and market capacity building, while ensuring that solutions are appropriate for Nepal's unique geographical and socioeconomic context.

⁹⁴ World Bank. 2024. *Program Document for the Third Nepal Finance for Growth Development Policy Credit Program*. <https://documents1.worldbank.org/curated/en/099041524091038418/pdf/BOSIB18634cbfa0931921a1566442b7ffb2.pdf>.

⁹⁵ Asia/Pacific Group on Money Laundering. 2023. *Anti-money laundering and counter-terrorist financing measures: Nepal Mutual Evaluation Report*. <https://www.nrb.org.np/contents/uploads/2023/09/Nepal-MER-2023.pdf>.

⁹⁶ *The Kathmandu Post*. 2025. It's official: Nepal falls back into FATF grey list. 22 February. <https://kathmandupost.com/national/2025/02/22/it-s-official-nepal-falls-back-into-fatf-grey-list>.

Table 1: Summary of Key Challenges

Category	Challenges
Infrastructure and Access	• Unreliable internet connectivity, particularly in rural and mountainous areas • Low affordability of smartphones, fixed broadband, and data services • Limited access to affordable and efficient cross-border payment options • Wide rural–urban disparities in internet usage and access • Unreliable middleware and payment gateway systems causing transaction errors • Incomplete interoperability across payment systems
Policy and Regulation	• Legacy regulatory frameworks with limited clarity for digital payments and fintechs • Supervisory gaps and uneven application of risk-based oversight • Absence of comprehensive cybersecurity, personal data protection, and consumer protection frameworks to regulate digital financial services • Restrictive foreign exchange regime with stringent approval processes that delay international transactions • Heightened Financial Action Task Force scrutiny and gray-listing risk undermining cross-border confidence and correspondent banking links
Market Readiness	• Low public awareness and trust in digital payment systems • Limited financial literacy and digital skills • High informality among micro, small, and medium-sized enterprises (MSMEs) limits access to formal banking, credit, and digital payment systems • High transaction fees and foreign exchange charges discourage MSMEs from adopting digital and cross-border payment options • Limited access to finance restricts MSMEs' ability to invest in digitalization and payment infrastructure • Limited technical capacity in MSMEs • Limited institutional support and capacity-building programs for MSME digital transformation
Technical and Operational Capacity	• Payment service providers must integrate individually with multiple systems rather than through a single unified interface, increasing cost and complexity • Legacy systems limit interoperability and hinder integration with international payment networks • Absence of standardized protocols, including messaging standards, security measures, and harmonized application programming interface specifications across payment systems • Weak implementation of cybersecurity measures despite existing guidelines • Shortage of skilled professionals in digital finance, payments technology, cybersecurity, data protection, and interoperability constrains payment system development and operational resilience

Source: Authors.

III. Learnings from Country-Level Digital Finance Initiatives

To effectively support Nepal in enhancing digital payments for trade facilitation, it is critical to examine global cases where digital payment systems have succeeded or encountered challenges, both domestically and in fostering cross-border transactions. The selected case studies come from both developing and developed countries that share relevant economic and infrastructural similarities with Nepal. These examples offer lessons on system design, stakeholder engagement, flexible regulatory alignment, strategies for international payment integration, and specific challenges smaller economies face. A key focus is on identifying models that have successfully facilitated cross-border trade and payments, providing a foundation for the recommendations that follow.

Highlights from Digital Payments Implementations Across the Globe

Kenya's M-Pesa

Before the launch of M-Pesa, Kenya faced significant barriers in financial inclusion, especially in rural areas where banking services were sparse. Most transactions relied heavily on cash, with limited access to secure financial services.⁹⁷ This cash-based economy hindered efficiency and increased the risk of theft and fraud. Against this backdrop, the regulatory authorities of Kenya looked to adopt mobile money banking that allowed people to access financial services even without a formal bank account. In 2007, Vodafone and Safaricom, the largest mobile network operator (MNO) in Kenya, launched M-Pesa. The following government actions supported the transformation of digital payment:

- **Regulatory support:** The Central Bank of Kenya adopted a facilitative regulatory approach, allowing Safaricom to operate M-Pesa under a letter of no objection, which enabled rapid deployment without the constraints of traditional banking regulations.
- **Financial inclusion initiatives:** The government promoted financial inclusion by supporting mobile money services, which provided a secure and efficient means for individuals to conduct transactions without traditional bank accounts. Originally designed with a simple and accessible interface using SMS, it was ideal for users with limited digital literacy, especially in rural regions.

⁹⁷ I. Mas and D. Radcliffe. 2010. Mobile Payments go Viral: M-Pesa in Kenya. *World Bank*. <https://documents1.worldbank.org/curated/en/638851468048259219/pdf/543380WP0M1PES1BOX0349405B01PUBLIC1.pdf>.

M-Pesa rapidly expanded, serving over 30 million active users monthly, as of 2022.⁹⁸ The country now boasts a 72% mobile money account penetration rate, a remarkable figure compared to neighboring countries such as Uganda, where only 43% of the population holds a mobile money account.⁹⁹ This growth has catalyzed trade, economic activity, and development in Kenya, empowering small businesses and rural communities. M-Pesa has evolved to include M-Pesa Global, which facilitates international remittances through partnerships with entities like Western Union and MoneyGram. M-Pesa Global now operates in other countries like Democratic Republic of the Congo, Egypt, Ethiopia, Ghana, Lesotho, Mozambique, and Tanzania, enabling smooth cross-border transactions and facilitating regional trade.¹⁰⁰

India's Unified Payments Interface

India's digital payments landscape was historically splintered among bank-proprietary apps, stand-alone digital wallets, and disparate QR code schemes. Cash remained the dominant mode of transaction, especially in rural areas, creating inefficiencies and friction in trade. This cash dependency limited financial inclusion, restricting the growth of small businesses. To address these issues, the Indian government, in collaboration with the Reserve Bank of India, established the National Payments Corporation of India (NPCI) in 2008. NPCI's mandate was to create a unified system, which led to the 2016 launch of the Unified Payments Interface (UPI), a real-time payment platform enabling instant, bank-to-bank transfers across various applications.¹⁰¹

India's national digital identity system called Aadhaar assigns a unique 12-digit identification number to Indian residents based on their biometric and demographic data. Aadhaar facilitates seamless digital identity verification and plays a critical role in enabling digital financial services, including UPI. The integration of Aadhaar with UPI streamlined customer onboarding processes and enhanced transaction security in India.

To promote UPI, the government implemented several incentives:

- **Subsidies and tax benefits:** Merchants were offered tax rebates, while customers received cash-back incentives, making UPI a cost-effective cash alternative.

Relevance to Nepal:

Kenya's approach highlights the significance of regulatory flexibility, which Nepal can emulate by streamlining the licensing framework and facilitating rapid innovation through regulatory sandboxes. Nepal can capitalize on its growing mobile connectivity by creating simplified, accessible mobile-based payment solutions, specifically targeting the informal economy, MSMEs, and remittance channels to boost financial inclusion and reduce reliance on informal transactions. Enhanced digital literacy programs, combined with a robust and reliable mobile network infrastructure, can accelerate adoption and effectiveness of such solutions across rural Nepal.

Moreover, M-Pesa's growth into M-Pesa Global shows that a strong domestic mobile money system, while initially focused on remittances, can become a platform for cross-border peer-to-peer transfers, potentially reducing costs and improving convenience compared to traditional channels. For Nepal, exploring how domestic payment capabilities can better integrate with international remittance services is a valuable lesson.

⁹⁸ K. Piper. 2020. What Kenya can teach its neighbors – and the US – about improving the lives of the “unbanked”. Vox. <https://www.vox.com/future-perfect/21420357/kenya-mobile-banking-unbanked-cellphone-money>.

⁹⁹ Financial Inclusion Insights. Kenya: Data at a glance. <https://finclusion.org/country/africa/kenya.html#dataAtAGlance> (accessed 30 October 2025).

¹⁰⁰ N. Wainaina. 2025. M-PESA's impact on digital payments beyond borders. *The Star*. <https://www.the-star.co.ke/news/big-read/2025-03-21-m-pesas-impact-on-digital-payments-beyond-borders>.

¹⁰¹ A. Mahesh and G. Bhat. 2021. Digital Payment Service in India – A Case Study of Unified Payment Interface. *International Journal of Case Studies in Business, IT, and Education*. 5 (1). pp. 256–265. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3885763.

- **Simplified regulatory frameworks:** To streamline payment service providers (PSPs) operations, the Reserve Bank of India simplified the regulatory framework by lowering license requirements and developing clear guidelines for payment aggregators and gateways.¹⁰²
- **Enhanced KYC for nonbank players:** UPI progressively strengthened its KYC framework to ensure consistent due diligence across both bank and nonbank participants. Initially, fintech apps and digital wallets could join UPI as third-party application providers if they partnered with a sponsoring bank, implemented NPCI's multi-factor authentication, and underwent periodic KYC audits using Aadhaar. In 2024, the Reserve Bank of India widened UPI services to include fully KYC-compliant digital wallets as pre-paid payment instruments.¹⁰³ These pre-paid payment instruments are required to perform in-person or video-based KYC verification, maintain tiered transaction limits, and notify NPCI of any onboarding anomalies. These measures align nonbank participants with bank-grade customer due diligence standards. UPI also deployed monitoring tools that use AI to detect unusual transaction patterns, further enhancing security, mitigating fraud risks, and fostering user trust.
- **Security protocols:** All UPI participants adhere to the Reserve Bank of India's Cyber Security Framework, mandating quarterly vulnerability assessments, incident-response drills, and third-party penetration testing.¹⁰⁴ Moreover, UPI's distributed ledger backbone logs every KYC lookup, ensuring auditability and consent-driven data exchange, while NPCI enforces strict least-privilege API-access tokens for nonbank apps.

These features and policies helped UPI reach over 9 billion monthly transactions by 2023, enhancing financial inclusion and reducing reliance on cash. UPI has also seen successful expansion into cross-border transactions. It established direct linkages with payment systems in other countries, a notable example being the UPI–PayNow linkage between India and Singapore, enabling instant, low-cost fund transfers between the two countries.¹⁰⁵ Furthermore, NPCI has actively worked to enable UPI acceptance at merchant locations in various countries, including Bhutan, France, Nepal and the United Arab Emirates. This typically involves allowing Indian travelers to pay using UPI apps via QR codes at merchant outlets, streamlining payments for tourism and small-value purchases. These linkages and acceptance networks simplify and reduce the cost of inward and outward remittances, increase speed and convenience for users, and produce direct benefits for local businesses (footnote 103).

Relevance to Nepal:

UPI's success in India underscores the transformative impact of interoperable payment systems. Given Nepal's current interoperability challenges, especially between mobile wallets and banks, adopting a unified interoperable standard similar to UPI can streamline transactions. Nepal can leverage UPI-like architecture to simplify cross-border transactions with India, its primary trading partner, thus addressing substantial inefficiencies in bilateral trade payments. Establishing a strong, standardized digital identity (KYC) system, as demonstrated by India's Aadhaar integration with UPI, can further facilitate seamless customer onboarding and enhance the security of digital transactions, addressing gaps in Nepal's AML/CFT compliance and identity verification challenges.

¹⁰² J. Asanova et al. 2024. Leveraging Lessons Learned from India's Unified Payments Interface for Digital Transformation in Asia and the Pacific. *ADB Briefs*. No. 299. <https://www.adb.org/publications/lessons-learned-india-unified-payments-interface>.

¹⁰³ Reserve Bank of India. 2024. Unified Payments Interface (UPI) access for Prepaid Payment Instruments (PPIs) through third-party applications. 27 December. <https://www.fidcindia.org.in/wp-content/uploads/2024/12/RBI-UPI-ACCESS-FOR-PPIs-27-12-24.pdf>.

¹⁰⁴ Government of India, Department of Financial Services. Cyber Security and Fintech. <https://financialservices.gov.in/beta/en/page/csft> (accessed 30 October 2025).

¹⁰⁵ NPCI International. What is the UPI–PayNow linkage? <https://www.npci.com/how-it-works/frequently-asked-questions/upipaynow> (accessed 30 October 2025).

Cambodia's Bakong

In 2020, the National Bank of Cambodia launched Bakong, a blockchain-based digital payment system designed to provide secure, real-time transactions within Cambodia's banking and payment ecosystem. Bakong also offers real-time cross-border payment capabilities.¹⁰⁶ Bakong's key features include the following:

- **Blockchain integration:** By incorporating blockchain technology, Bakong enables secure, decentralized, and real-time transactions. Utilizing a permissioned blockchain technology, its transparency and security protocols help build trust, minimize fraud, and support compliance with regulatory requirements.
- **Interoperability across financial institutions:** Bakong connects banks and PSPs under a single platform, simplifying digital payments across institutions and reducing transaction costs. This integration allows users to transfer funds across banks seamlessly, supporting small businesses and easing daily transactions.
- **Cross-border remittance support:** Recognizing the importance of migrant remittances to Cambodia's economy, the National Bank of Cambodia partnered with foreign banks, particularly in Malaysia and Thailand. By connecting Bakong with partner country networks, transfers became more direct and transparent, reducing intermediaries, transaction fees, and settlement times for Cambodians.

By 2024, the number of customers with bank accounts or digital wallets that are capable of conducting transactions via Bakong reached 30 million accounts, which is equivalent to 1.69 times the total population.¹⁰⁷ In addition, Bakong facilitates cross-border QR code payment, benefiting tourists and small traders. For example, Cambodians can scan Thai QR codes for payments in Thailand using their Bakong app. Between 2020 and 2023, cross-border QR payment transactions between Cambodia and Thailand reached B27 million (\$830,000).¹⁰⁸ As of 2025, Cambodia signed cross-border payment cooperation agreements with regional peers such as the Lao People's Democratic Republic, Malaysia, Thailand, and Viet Nam, as well as with the People's Republic of China and the Republic of Korea.¹⁰⁹

Relevance to Nepal:

Cambodia's experience with Bakong highlights the importance of central bank-driven innovation in integrating digital payment systems across diverse banking institutions. For Nepal, Bakong offers a model for addressing its technical integration challenges due to the fragmented banking infrastructure. Implementing a central bank-backed national digital payment platform in Nepal can improve the efficiency and reliability of cross-border transactions, simplify regulatory compliance, and enhance payment gateway stability.

¹⁰⁶ M. Takemiya. 2021. Cambodia's digital currency can show other central banks the way. *World Economic Forum*. 30 August. <https://www.weforum.org/stories/2021/08/cambodias-digital-currency-ishowing-other-central-banks-the-way/>.

¹⁰⁷ H. Pisei. 2025. NBC report strong uptake of Bakong digital transfer system. *The Phnom Penh Post*. 26 February 2025. <https://www.phnompenhpost.com/business/nbc-report-strong-uptake-of-bakong-digital-transfer-system>.

¹⁰⁸ Aclada Bank. 2023. Cross-Border QR Payment Between Cambodia and Thailand Reaches KHR 52 Million Per Day. 6 June. https://www.acladabank.com.kh/kh/eng/md_In20230606.

¹⁰⁹ S. Ravindran. 2025. Bilateral Cooperation on Cross-Border Payment and Remittance. *Khmer Times*. 20 March. <https://www.khmertimeskh.com/501657303/bilateral-cooperation-on-cross-border-payment-and-remittance/>.

Brazil's Pix

Launched in 2020 by the Central Bank of Brazil, Pix is a real-time payment system.¹¹⁰ Through Pix, individuals and businesses can transfer funds instantly using mobile devices, QR codes, or unique identifiers such as email addresses. Designed with accessibility, affordability, and integration in mind, Pix has reshaped Brazil's financial landscape by enabling secure, fast, and cost-effective digital transactions across various platforms. Key features of Pix include the following:

- **Integrated system design:** Pix's real-time payment system is well integrated across banks, digital wallets, and fintech applications.
- **Cost-effectiveness:** Pix transactions are free for individuals, while businesses are charged a minimal fee. This approach has reduced barriers to digital payment usage, especially among low-income users who may be deterred by transaction fees.
- **Integration with public sector transactions:** A notable strength of Pix is its integration into Brazil's public sector, where it has been utilized for government disbursements, tax collections, and other public payments. This integration showcases how governments can also benefit from well-designed digital payment methods.

By 2024, Pix achieved remarkable penetration, with over 152 million of Brazil's 163 million adults registered, and over 201 million transactions processed daily.¹¹¹ Initially developed for domestic transactions, the Central Bank of Brazil is now exploring partnerships to link Pix with international payment systems, aiming to facilitate cross-border transactions and strengthen Brazil's economic connectivity (footnote 111).

Pakistan's Mobile Money Platforms, Led by Mobile Network Operators

Pakistan is a notable example of how a flexible regulatory approach can enable MNOs to drive financial inclusion and digital payment adoption. Recognizing the vast reach of mobile telephony compared to traditional banking infrastructure, the State Bank of Pakistan established frameworks that allowed MNOs to operate as PSPs and offer a range of financial services, often through their mobile money platforms like Easypaisa (Telenor) and JazzCash (Jazz). Key enablers and outcomes include the following:

- **Enabling regulatory framework:** The State Bank of Pakistan introduced regulations for branchless banking and electronic money institutions, which paved the way for MNOs to offer financial services.¹¹² This included tiered KYC requirements and frameworks, making it easier for unbanked individuals to open basic mobile wallets. This has been a key enabler for expanding access, especially in

Relevance to Nepal:

Brazil's Pix highlights the value of real-time, low-cost payments in promoting widespread digital payment adoption. Nepal faces challenges with high transaction costs, especially in cross-border payments. Introducing a Pix-like instant payment system can reduce costs and processing times, thereby encouraging MSMEs and rural populations to embrace digital payments. Clear, unified regulatory guidance and cybersecurity protocols are critical lessons from Pix, aligning closely with Nepal's identified regulatory gaps in cybersecurity preparedness and consumer protection frameworks.

¹¹⁰ Payments and Commerce Market Intelligence. 2025. *Pix in Brazil: What to Expect in 2025 and beyond*. <https://paymentscmi.com/insights/pix-in-brazil-latest-statistics-central-bank/>.

¹¹¹ M. Ceurvels. 2024. Pix Explainer 2024: How Brazil's Instant Payment System Is Revolutionizing Financial Services. *emarketer*. 9 February. <https://www.emarketer.com/content/pix-explainer-2024>.

¹¹² State Bank of Pakistan. 2024. *National Financial Inclusion Strategy 2024–28*. <https://www.sbp.org.pk/nfis/nfis2024-28.pdf>.

rural and underserved areas.¹¹³ The State Bank of Pakistan also worked closely with the Pakistan Telecommunication Authority to develop unified regulations that allow MNOs to participate in mobile banking, using both telecom-led and bank-led models.¹¹⁴

- **Diverse service offerings by MNOs:** MNO-led platforms became major channels for paying utility bills, leveraging the convenience of mobile top-up locations as payment points or direct payments from mobile wallets. In addition, mobile money services enabled easy and affordable domestic remittances and increased person-to-person transfers, crucial in a country with significant internal migration. The government also began utilizing these platforms for social welfare disbursements, ensuring transparent and efficient delivery of government-to-person funds. Leveraging flexible insurance regulations, MNOs were able to act as agents and develop insurance and other financial products.¹¹⁵ They distributed micro-insurance products, which are often bundled with mobile services or offered via mobile wallets, expanding access to risk mitigation tools for low-income populations.
- **Use of existing technology:** The model leveraged existing mobile infrastructure—using Unstructured Supplementary Service Data (USSD) for basic phones and mobile apps for smartphones—and the country’s extensive mobile subscriber base, minimizing the need for new infrastructure for access to basic financial services.

MNO-led services increased the percentage of adults with a financial account in Pakistan. For instance, the number of depositors in Pakistan grew from 54 million in 2018 to 88 million in 2023, with a notable increase in women’s financial inclusion (from 23% to 47% of women having accounts).¹¹⁶ The extensive agent network, established by MNOs and primarily made up of small retailers, became crucial access points for cash-in/cash-out transactions and account opening, especially in rural and underserved areas.

Relevance to Nepal:

Pakistan’s case study underscores the importance of a flexible regulatory framework co-developed by the central bank and telecommunication authority. Strategies that Nepal can adopt to strengthen its digital payment ecosystem include empowering MNOs, simplifying account opening with tiered KYC requirements, and promoting a diverse range of services.

Nigeria’s eNaira

In 2021, the Central Bank of Nigeria launched eNaira, Africa’s first Central Bank Digital Currency (CBDC). Designed as both a medium of transaction and a store of value, eNaira was introduced with the goals of enhancing financial inclusion and simplifying cross-border trade. Despite these ambitious goals, the adoption rate within the first year was low, with fewer than 0.5% of Nigerians engaging with the platform.¹¹⁷ This limited uptake was attributed to a combination of technical challenges, public distrust, and inadequate infrastructure. Key challenges from eNaira’s launch are often attributed to the following factors:¹¹⁸

¹¹³ E. Shahzad and R. Arif. 2024. The Role of Mobile Wallets in Fintech: Enhancing Financial Independence for Women in Pakistan. *Mahbub ul Haq Research Centre at LUMS*. 1 March. <https://mhrc.lums.edu.pk/role-mobile-wallets-fintech-enhancing-financial-independence-women-pakistan>.

¹¹⁴ Pakistan Telecommunication Authority. 2023. *PTA and SBP to formulate Third Party Mobile Banking Regulations*. <https://www.pta.gov.pk/category/pta-and-sbp-to-formulate-third-party-mobile-banking-regulations-137813870-2023-06-01>.

¹¹⁵ GSMA. 2014. *Promising Starts in Mobile Microinsurance: Tigo Senegal & Telenor Pakistan*. <https://www.gsma.com/solutions-and-impact/connectivity-for-good/mobile-for-development/wp-content/uploads/2014/02/Promising-Starts-in-Mobile-Microinsurance.pdf>.

¹¹⁶ M. Aazim. 2025. Finance: Slowly fixing the economy. *Dawn*. 14 July. <https://www.dawn.com/news/1924048>.

¹¹⁷ N. Anthony. 2022. Nigeria Restricts Cash to Push Central Bank Digital Currency. *CATO Institute*. 19 December. <https://www.cato.org/blog/central-bank-digital-currency-war-cash>.

¹¹⁸ E. Monye. 2024. Why Nigeria’s Controversial Naira Redesign Policy Hasn’t Met Its Objectives. *Carnegie Endowment for International Peace*. 18 January. <https://carnegieendowment.org/research/2024/01/why-nigerias-controversial-naira-redesign-policy-hasnt-met-its-objectives?lang=en>.

- **Technical issues and user experience:** Shortly after its release, eNaira faced technical glitches that impacted user experience. These issues hindered the platform's accessibility and reliability, causing frustration among early adopters and deepening skepticism around the technology. This case underscores the importance of thoroughly testing digital payment platforms before a full-scale launch. Implementing sandbox environments or pilot programs can help address potential issues, enabling a smoother rollout and building early confidence in the system.
- **Public trust and digital literacy:** eNaira's adoption was further hindered by low levels of public trust in digital currencies and a limited understanding of how CBDCs function. In Nigeria, as in many regions, concerns about data privacy and security create resistance to new digital financial tools. This highlights the need for nationwide public education campaigns and transparent communication strategies when introducing innovative financial technology. By actively engaging the public, addressing concerns, and providing clear, accessible information on digital currency benefits and safeguards, governments can build the trust necessary for adoption.¹¹⁹
- **Infrastructure challenges:** Nigeria's digital infrastructure posed another barrier, limiting eNaira's reach and usability, especially in rural areas. Reliable internet connectivity and robust digital infrastructure are essential for the success of digital currencies and payment platforms.

Sri Lanka's Domestic Digital Progress and Cross-Border Payment Hurdles

Sri Lanka has made notable strides in its domestic digital financial services landscape, partly driven by an enabling regulatory environment from the Central Bank of Sri Lanka that allowed MNOs and other nonbanks to participate actively. MNO-led platforms like Dialog's eZ Cash and Mobitel's mCash have been instrumental in expanding access to basic financial services, offering services such as mobile money, utility bill payments, domestic peer-to-peer transfers, and local merchant payments.¹²⁰

Despite these domestic achievements and a relatively well-banked and digitally literate population, Sri Lanka faces challenges in its integration with the global digital economy, particularly concerning international PSPs. For example, services like PayPal, while allowing Sri Lankans to make outbound payments, have not offered inbound payment functionality for receiving payments.¹²¹ This restriction is attributed to Sri Lanka's regulatory constraints and commercial dynamics, including local banks' protection of their foreign exchange remittance revenues and resistance to changes that would allow nonbank platforms to compete directly. Some analysts point to Sri Lanka's relatively small market size and perceived low transaction volumes as factors that may discourage international PSPs from investing in full payment functionalities.¹²² These limitations have considerable implications:

Relevance to Nepal:

Nigeria's eNaira demonstrates the complexities involved in introducing CBDCs but also underscores their potential in reducing dependency on cash and improving cross-border payment efficiency. As Nepal is actively exploring its own CBDC, it can learn from Nigeria's challenges by prioritizing cybersecurity measures, clear regulatory frameworks, and public awareness initiatives. Addressing infrastructure readiness, particularly internet penetration and reliability, remains a crucial priority for successful CBDC implementation in Nepal.

¹¹⁹ A. Osae-Brown, M. Fatunde, and R. Olurounbi. 2022. Digital-Currency Plan Falters as Nigerians Defiant on Crypto. *Bloomberg*. <https://www.bloomberg.com/news/articles/2022-10-25/shunned-digital-currency-looks-for-street-credibility-in-nigeria>.

¹²⁰ U. Jayasinghe. 2024. Sri Lankan FinTech Landscape for Financial and Non-Financial Services and Prospects. *APABI Focus*. 2 October. [https://web.tabf.org.tw/Page/EDMimg/APABI/2024_10/Sri%20Lankan%20FinTech%20Landscape%20for%20Financial%20and%20Non-Financial%20Services%20\(1BSL\).pdf](https://web.tabf.org.tw/Page/EDMimg/APABI/2024_10/Sri%20Lankan%20FinTech%20Landscape%20for%20Financial%20and%20Non-Financial%20Services%20(1BSL).pdf).

¹²¹ *Lanka e News*. 2024. Sri Lanka and PayPal: A Love Affair That's Only Half a Relationship! 15 December. <https://www.lankaenews.com/news/4029/en>.

¹²² *Colombo Initiative*. 2021. Permitting Foreign Currency Inward Remittances through Online Platforms: Challenges and Solutions. 24 October. <https://web.archive.org/web/20211028112053/https://colomboinitiative.org/permitting-foreign-currency-inward-remittances-through-online-platforms-challenges-and-solutions/>.

- **Hindrance to global e-commerce and freelancing:** Sri Lankan freelancers, service exporters, and small e-commerce businesses find it difficult to receive payments from international clients seamlessly and cost-effectively. They often have to resort to more expensive or less convenient methods like international bank transfers with higher fees and less favorable exchange rates, or alternative platforms that might not be universally accepted by their clients.
- **Barriers to digital economy participation:** The inability to easily receive international digital payments limits the potential for Sri Lankans to fully participate in and benefit from the global digital economy, impacting export earnings from digital services.

Relevance to Nepal:

Sri Lanka's experience shows that achieving domestic digital payment success does not guarantee full integration into the global digital economy, providing a critical lesson for Nepal. While Nepal's initial focus may be on developing its domestic digital payment ecosystem, forward-looking policies and strategic partnerships that will enable seamless cross-border transactions are necessary to avoid isolating its own growing digital economy and undermining trade facilitation goals.

Analysis of Success Factors and Challenges: Lessons for Nepal

The case studies presented in Chapter III, A reveal several critical success factors that Nepal can draw from to improve its digital payment ecosystem. These are discussed below and summarized in Table 2.

Robust and Reliable Digital Infrastructure

The foundation of any successful digital payment system, whether domestic or cross-border, is robust and reliable infrastructure. The success of Brazil's Pix and the initial struggles of Nigeria's eNaira demonstrate how infrastructure reliability directly impacts adoption rates. For cross-border payments, India's UPI leverages its strong domestic infrastructure to build international linkages. Cambodia's Bakong, while innovative with blockchain, also relies on underlying connectivity to function effectively for both domestic and cross-border transactions. Kenya's M-Pesa demonstrates that even simple SMS-based systems require widespread and reliable mobile network coverage, which later served as a foundation for M-Pesa Global remittances. For Nepal, specific lessons include the following:

- **Network coverage and reliability:** Given Nepal's challenges in rural connectivity and maintaining service quality, prioritizing investment in expanding and strengthening the resilience of its mobile network infrastructure will be critical. Enhancing internet penetration rates and quality will improve user experiences, encourage adoption, and support transaction reliability.
- **Adaptable technology solutions:** Cambodia's Bakong illustrates the potential for blockchain technology to support reliable financial services, even with intermittent connectivity. Implementing such decentralized and resilient technological solutions in Nepal can mitigate geographical and infrastructural limitations, and contribute to transaction integrity and reliability across diverse regions.

Regulatory Support and Flexibility

Kenya's M-Pesa thrived due to a facilitative test-and-learn approach from the Central Bank of Kenya. Pakistan and Sri Lanka demonstrated how enabling regulations for nonbanks, especially MNOs, can drive domestic financial inclusion, bill payments, and the provision of services like micro-insurance through mobile channels. India's UPI success was backed by strong regulatory push from the Reserve Bank of India and government

initiatives. For cross-border payments, successful integrations like India–Singapore UPI linkage or Cambodia–Thailand QR interoperability often depend on clear bilateral regulatory agreements and harmonized standards. The analysis reveals three critical success factors:

- **Regulatory agility and sandboxes:** Kenya’s regulatory flexibility allowed M-Pesa to rapidly innovate and scale. Nepal can similarly establish regulatory sandboxes that facilitate experimentation with innovative digital payment solutions under a controlled environment, promoting quick adaptation and responsiveness to market needs.
- **Robust digital identity systems:** India’s Aadhaar system demonstrates how a national digital identity framework can streamline customer onboarding, reduce transaction fraud, and facilitate broader financial inclusion. Nepal can develop a similarly robust digital identity system to simplify KYC procedures, enhance payment security, and support inclusive access to digital financial services, particularly benefiting informal sector participants and underserved communities.
- **Cross-border regulatory coherence:** Cambodia’s Bakong and Brazil’s Pix underscore clear regulatory frameworks as essential for enabling effective cross-border transactions. Given Nepal’s extensive trade and financial ties with India and its ambition to expand regional trade, establishing transparent and coherent regulatory frameworks that simplify international payment procedures is crucial.

Interoperability and Standardization

Interoperability and standardization are vital for integrating multiple payment systems and ensuring cross-border compatibility. Moving forward, Nepal needs to consider the following:

- **Unified technical standards:** India’s UPI demonstrates the importance of standardized technical protocols, allowing seamless transactions across diverse banks and PSPs. For Nepal, adopting and enforcing standardized protocols will reduce fragmentation, enabling efficient and frictionless domestic and international digital transactions.
- **Cross-border interoperability:** Cambodia’s Bakong employs blockchain technology for secure, efficient cross-border payment transactions. By adopting similar interoperable solutions, Nepal can simplify and expedite cross-border payments, especially with key trading partners like India. This approach can reduce operational and transaction costs in international trade.
- **Integrated QR payment standardization:** India’s UPI excelled by enabling transactions across different banks and payment apps through standardized QR codes. For Nepal, where QR payments are expanding rapidly, but interoperability challenges persist, introducing a standardized QR code system can simplify merchant adoption, encourage customer use, and accelerate digital payments acceptance nationwide.

User-Friendly and Accessible Interfaces

High adoption rates are often linked to user-friendly interfaces and clear value propositions. Brazil’s Pix gained traction with simple payment aliases. Kenya’s M-Pesa succeeded with an SMS-based interface accessible on basic phones. The MNO-led services in Pakistan and Sri Lanka became popular by offering convenient solutions for high-frequency needs like utility bill payments and domestic remittances. The analysis emphasizes that intuitive, user-centric payment interfaces increase digital payment adoption:

- **Intuitive payment experience:** Digital payment solutions need to be simple, intuitive, designed with users in mind, and accessible to individuals across all demographic groups and levels of digital literacy.
- **Multichannel payment options:** Offering multiple simple and accessible payment channels, including SMS, smartphone apps, and QR codes, can enhance adoption and convenience in Nepal’s diverse consumer base.

Public Trust and Digital Literacy

Public trust and digital literacy are foundational to sustainable digital payment adoption. Building trust requires ensuring system security, personal data protection, and providing consumer protection and dispute resolution mechanisms. Nigeria's eNaira faced trust issues, impacting adoption. Successful systems like UPI have strong security protocols and clear redressal mechanisms. Comprehensive digital literacy programs are also essential, as highlighted by challenges in various countries if adoption is pushed without adequate user education. Areas that Nepal needs to prioritize include the following:

- **Building trust:** Focusing on cybersecurity measures, system stability, and clear communication about the safety and reliability of digital payments can build sustained public trust.
- **Improving digital and financial literacy:** Nepal needs to develop digital and financial literacy programs that raise awareness of digital finance benefits, safe transaction practices, and cybersecurity, targeting MSMEs, rural populations, and underserved groups.
- **Incentivizing adoption through economic benefits:** Since Nepal Rastra Bank (NRB) has been successful in formalizing remittance transfers through cash and other incentives already, Nepal can consider similar incentivizing policies, strategically designed to encourage rural MSMEs and merchants to shift from cash to digital transactions, creating immediate economic benefits and driving sustained adoption.

Table 2: Summary of Critical Success Factors

Critical Success Factor	Recommendations for Nepal
Robust and Reliable Digital Infrastructure	• Prioritize improving mobile network infrastructure to bridge rural connectivity gaps • Adopt resilient technologies like blockchain
Regulatory Support and Flexibility	• Establish regulatory sandboxes for innovative digital payment trials • Consider engaging mobile network operators in the domestic payment space • Develop a robust digital identity framework to streamline user onboarding and enhance transaction security • Define clear regulations to streamline and facilitate cross-border payments, especially with India
Interoperability and Standardization	• Adopt unified payment standards to enhance domestic and international interoperability • Utilize interoperable blockchain-based platforms for efficient cross-border payments • Introduce standardized Quick Response payment systems nationwide to enhance merchant and user convenience • Actively participate in regional payment platform development
User-Friendly and Accessible Interfaces	• Create intuitive interfaces suitable for diverse user groups, minimizing technological barriers • Offer multiple low-tech payment options, like SMS, to ensure broad user inclusion • Implement targeted merchant incentives and streamlined onboarding processes for faster adoption
Public Trust and Digital Literacy	• Strengthen cybersecurity and ensure transparent communication • Expand digital literacy programs, targeting micro, small, and medium-sized enterprises, and rural and underserved communities • Provide economic incentives, such as reduced fees and tax benefits, to encourage broader digital payment adoption

Source: Authors.

IV. Recommendations

Drawing on the analysis of Nepal's digital payment ecosystem and lessons from implementations across the globe, this section presents strategic recommendations for Nepal. The recommendations address both trade facilitation priorities and broader system development needs, recognizing their distinct requirements while acknowledging their interconnections.

Trade Facilitation Priorities

Because cross-border payment integration depends on the reliability, reach, and adoption of the domestic payment ecosystem, the recommendations are distinguished between direct trade facilitation priorities and enabling domestic reforms.

- **Expedite the resolution of commission structure issues for Nepal–India QR payments:** Engage directly with the Reserve Bank of India and NPCI to establish a sustainable framework that enables the already-developed reciprocal functionality for Nepalis in India.
- **Implement dedicated payment corridors with key trading partners:** Create specialized payment channels with Bangladesh, the People's Republic of China, and India with enhanced reliability. Pursue regional cooperation through SASEC to standardize cross-border payment procedures.
- **Deploy specialized payment solutions for the tourism and remittance sectors:** Develop targeted solutions for these economically vital sectors with multicurrency processing capabilities and streamlined onboarding for tourism businesses and remittance service providers.
- **Enhance reliability of international payment gateways:** Address middleware issues connecting third-party payment gateways to improve system stability. Establish redundant gateway infrastructure with automatic failover capabilities to maintain operations during system outages.

Regulatory Alignment and Compliance

- **Strengthen AML/CFT framework to address FATF requirements:** Implement comprehensive reforms to exit the FATF gray list, focusing on enhanced monitoring systems, transparent beneficial ownership, and improved reporting mechanisms.
- **Implement risk-based KYC procedures:** Building on the existing tiered model, create simplified KYC processes for low-value transactions while maintaining robust verification for higher-risk activities.
- **Harmonize foreign exchange regulations with trading partners:** Work with trading partners to develop standardized foreign exchange procedures, addressing challenges posed by Nepal's currency peg with the Indian rupee.
- **Develop and enact a comprehensive Personal Data Protection Act** aligned with international standards (e.g., European Union's General Data Protection Regulation), moving beyond the existing Privacy Act 2018.

- **Formalize and expand the Financial Consumer Protection Framework**, creating clear regulations for digital financial services, especially for cross-border transactions.
- **Create a regulatory framework for e-commerce payments**: Create clear guidelines for e-commerce transactions, addressing consumer protection, merchant liability, and dispute resolution mechanisms.

Payment System Interoperability

- **Finalize and mandate the planned open-API rule book for trade finance and cross-border payments**: This will support the development of common APIs connecting banks, businesses, and government agencies with consistent data formats and security protocols.
- **Create standardized QR payment framework**: Develop a unified QR code standard (expanding NEPALPAY QR adoption) to address current fragmentation and enable cross-border QR payments with key trading partners, particularly India.
- **Establish interoperability between mobile wallets and banking systems**: Enforce technical standards that eliminate the need for wallet-to-wallet transactions to be routed through banks, reducing complexity and costs.

Trade-Focused Market Development

- **Introduce incentive programs for export-oriented businesses**: Develop support packages, including transaction fee subsidies and tax incentives for digital payment adoption, with greater benefits for MSMEs engaged in cross-border trade.
- **Develop specialized digital solutions for informal sector integration**: Create simplified digital payment tools for Nepal's informal sector with minimal documentation requirements and appropriate transaction limits.
- **Implement digital documentation solutions for cross-border trade**: Develop blockchain-based systems for secure, tamper-proof trade documentation linked to payment flows, reducing administrative burdens and processing times.
- **Establish cross-border fintech partnerships**: Foster relationships between Nepali PSPs and international fintech companies to adapt proven global solutions to Nepal's specific needs.

Broader Digital Payment System Recommendations

Domestic Infrastructure Enhancement

- **Deploy low-bandwidth and offline-capable payment solutions**: Develop SMS-based and USSD payment options for areas with limited connectivity. These solutions should support basic transaction types on feature phones with intermittent connections.
- **Implement hybrid connectivity models for rural coverage**: Combine fiber-optic networks with alternative technologies to reach rural areas, prioritizing payment traffic on available bandwidth and developing offline transaction capabilities.
- **Support the effective implementation of the Cyber Resilience Guidelines**: Provide technical assistance and clear enforcement pathways to help financial institutions meet compliance requirements. Establish a national Financial Sector Computer Security Incident Response Team.
- **Establish robust monitoring systems**: Implement real-time monitoring platforms providing visibility into system performance, with automated alerts and performance dashboards to quickly identify and resolve issues.

Market Development and Capacity Initiatives

- **Launch targeted merchant adoption programs:** Create support packages with subsidized equipment, fee waivers, and training resources, particularly for MSMEs.
- **Develop a nationwide network of cash-in/cash-out agents to ensure last-mile financial inclusion:** This would involve creating clear regulatory guidelines and incentives for agents (distinct from merchants) to serve as a critical bridge between the cash and digital economies.
- **Create a public-private partnership framework:** Incentivize private sector investment in expanding access to digital financial services and agent networks in underserved regions.
- **Develop comprehensive digital and financial literacy programs:** Implement training initiatives with specialized modules for different user groups, including hands-on sessions and multilingual materials accessible through multiple channels.
- **Address skill shortages through partnerships:** Collaborate with educational and training institutions to develop specialized fintech curriculum and certification programs to address the digital talent shortage exacerbated by brain drain.
- **Create a formal regulatory sandbox, as planned by NRB,** to foster controlled testing of new financial technologies.
- **Promote competition and innovation** by clarifying the regulatory distinction between banking and payment services, and facilitating fair infrastructure access for licensed nonbank PSPs.

High-Level Implementation Road Map

Table 3 outlines a high-level implementation road map with milestones over a 5-year horizon. Timelines are indicative and depend on resource allocation and stakeholder commitment. The full list of stakeholders mentioned is given in Appendix 2.

Table 3: High-Level Implementation Road Map

Phase	Focus and Expected Outcome	Key Milestones and Actions	Responsible Stakeholders
Phase 1: Foundations Immediate and ongoing actions through mid-2026	Focus: Stabilize infrastructure, establish core regulatory frameworks, address urgent compliance issues, and set standards Expected outcomes: Financial Action Task Force (FATF) compliance progress, improved domestic system stability, foundational legal certainty established, and resolution of a key cross-border payment bottleneck	Implement comprehensive anti-money laundering/combating the financing of terrorism reforms to address FATF requirements	FIU-Nepal, NRB, MOF
		Draft comprehensive Personal Data Protection Act and consumer protection regulations for digital finance	MOCIT, NRB, DOCSCP
		Implement core domestic payment gateway redundancy and establish the Financial Sector Computer Security Incident Response Team	NRB, NCHL, National Cyber Security Center

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Table 3 *continued*

Phase	Focus and Expected Outcome	Key Milestones and Actions	Responsible Stakeholders
		Resolve Nepal–India Quick Response (QR) payment commission structure to enable reciprocal functionality	NRB, NCHL, NPCI International Payments Limited
		Publish the planned open application programming interface rule book and mandate ISO 20022	NRB, NBSM
		Develop regulatory framework and incentive structure for a national cash-in/cash-out agent network	NRB, MOF
Phase 2: Scaling and Integration Mid-2026–mid-2028	Focus: Implement regulations, expand reliable infrastructure, drive adoption, enhance interoperability, and pilot innovations Expected outcomes: Foundational regulations in force, significantly improved infrastructure reliability, increased user and merchant adoption, and enhanced domestic and initial cross-border interoperability	Enact and begin enforcement of the Personal Data Protection Act and consumer protection regulations	Government, MOCIT, DOCSCP
		Launch nationwide merchant adoption programs and targeted support and incentives for micro, small, and medium-sized enterprises	NRB, MOICS, Banks/ PSPs
		Launch and scale the national cash-in/cash-out agent network, focusing on rural and underserved areas	Banks, payment service providers, telecommunication operators
		Establish a public–private partnership framework to incentivize private sector digital financial service expansion into underserved regions	NRB, MOCIT, MOF
		Mandate and enforce standardized open banking APIs and enhance NEPALPAY QR for international interoperability	NRB, NCHL, NBSM
		Pilot the national digital identity system for streamlined know-your-customer processes	Department of National Identification and Civil Registration, NRB

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Table 3 *continued*

Phase	Focus and Expected Outcome	Key Milestones and Actions	Responsible Stakeholders
Phase 3: Optimization and Innovation Mid-2028–mid-2030	Focus: Refine systems for cost-efficiency, deepen regional integration, foster innovation, and scale successful pilots Expected outcomes: Optimized and cost-effective digital payment ecosystem supporting seamless regional trade, and mature and competitive fintech market	Focus on initiatives to drive down cross-border transaction costs based on Phase 2 efficiencies	NRB, market participants
		Deepen integration with regional and global payment networks (e.g., South Asia Subregional Economic Cooperation)	NRB, MOICS, Ministry of Foreign Affairs
		Scale successful pilots (e.g., Central Bank Digital Currency, blockchain for trade) based on outcomes and established frameworks	NRB, MOICS
		Foster advanced fintech partnerships for innovative solutions	NRB, fintech community
		Continuously refine regulatory frameworks based on market developments and data analysis	NRB, MOCIT, DOCSCP

DOCSCP = Department of Commerce, Supplies and Consumer Protection; FIU-Nepal = Financial Intelligence Unit; MOCIT = Ministry of Communications and Information Technology; MOF = Ministry of Finance; MOICS = Ministry of Industry, Commerce, and Supplies; NBSM = Nepal Bureau of Standards and Metrology; NCHL = Nepal Clearing House Limited; NRB = Nepal Rastra Bank; PSPs = payment service providers.

Source: Authors.

Stakeholder Coordination and Monitoring

- **Establish a high-level National Digital Payments Steering Committee** comprising senior representatives from NRB, MOCIT, Ministry of Finance, Ministry of Industry, Commerce, and Supplies, NCHL, financial institutions, licensed PSOs and PSPs, telecommunication operators, and key business associations (e.g., Federation of Nepalese Chambers of Commerce and Industry). This committee should meet regularly to oversee strategy implementation, ensure inter-agency coordination, resolve bottlenecks, and adapt plans as needed. Responsibility for the operation, security, and management of digital payments infrastructure is proposed to be handled by NRB. The Payments Steering Committee will establish and sponsor coordination mechanisms with the Department of Information Technology for Digital Standards and with the National Cyber Security Center for Cyber Security Standards.
- **Foster regular dialogue and feedback mechanisms** between regulators, industry players, and end users (consumers and businesses) through workshops, consultations, and surveys.
- **Utilize a dedicated dashboard or monitoring framework** to track progress against the road map's milestones and success metrics, enabling data-driven adjustments and reporting.

Success Metrics for Trade Facilitation

- **Successfully exit the FATF gray list within 2 years:** This is the most critical near-term metric, as remaining on the gray list threatens correspondent banking relationships that underpin all international trade and finance.
- **Achieve sustained 99.5% uptime for international payment systems by end of 2027:** Measure through independent monitoring systems with transparent reporting of scheduled and unscheduled downtime. The end of 2027 time frame reflects the significant infrastructure upgrades, operational process improvements, and testing required to achieve and sustain this high level of international system reliability (aligns with completion of Phase 2 actions).
- **Reduce average end-to-end cross-border payment processing time by 50% by end of 2027:** Track the complete payment life cycle from initiation to settlement with progressive improvement targets. This timeline aligns with the implementation of foundational changes from Phases 1 and 2, such as API standardization and streamlined regulations.
- **Increase digital payment adoption among export and import businesses to 60% by end of 2027:** Conduct regular surveys of businesses engaged in international trade to measure adoption and identify barriers. An end of 2027 goal provides the necessary period for building trust, rolling out support programs, and allowing businesses time to adapt following the foundational improvements in Phases 1 and 2.
- **Double the volume and value of digital cross-border transactions (excluding large informal flows) by 2028:** Monitor transaction volumes, values, and growth rates across different payment channels and corridors. This target reflects the cumulative impact of improved systems, lower costs, and increased user adoption achieved through the initial road map phases.

The successful enhancement of Nepal's digital payment ecosystem requires balanced attention to both trade facilitation and domestic system development. Implementation needs to be guided by adaptive management, data-driven insights, regular stakeholder feedback, and strong political will. Critical to success will be effective coordination among relevant stakeholders.

APPENDIX 1

Stakeholder Interviews

In preparing this report, the research team conducted five interviews with relevant stakeholders in Nepal's payment ecosystem to accurately capture on-the-ground circumstances. These interviews were primarily conducted virtually, but some were carried out in written form.

Table A1: List of Stakeholder Interviews

Category	Affiliation	Representatives
Government	Nepal Rastra Bank	Payment Systems Department
Financial infrastructure provider	Nepal Clearing House Limited	Payment System
Payment service providers	Fonepay / F1Soft Namaste Pay / Nepal Digital Payments Company	Not disclosed
Technical expert	Kathmandu Fintek	Sanjib Subba

Source: Authors.

APPENDIX 2

Primary Stakeholders in Nepal's Digital Payment Ecosystem

The main stakeholders and their roles in Nepal's digital payment ecosystem are provided in Table A2.

Table A2: List of Primary Stakeholders in Nepal's Digital Payment Ecosystem

Category	Stakeholder	Roles
Ministries	Ministry of Industry, Commerce, and Supplies (MOICS)	Sets national trade policy, leads negotiations under frameworks such as the World Trade Organization, South Asian Free Trade Area, and Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation, and oversees export schemes. Its role is key for aligning e-trade documentation with cross-border payment systems The Department of Commerce, Supplies, and Consumer Protection enforces consumer protection and competition rules in e-commerce and digital payments
	Ministry of Finance (MOF)	Sets tariff policies and fiscal incentives, implemented by the Department of Customs The Department of Customs operates the ASYCUDA World system, sets customs procedures, and collects duties
	Ministry of Communications and Information Technology (MOCIT)	Leads the Digital Nepal Framework, spectrum policy, and government data center strategy—critical for the connectivity and cloud hosting that digital payment systems need The Department of Information Technology oversees government cloud and data center operations, and manages the digital infrastructure and application programming interface (API) gateways
	National Planning Commission	Incorporates digital payment and trade facilitation targets into five-year plans, allocates budget ceilings, and monitors alignment with the Sustainable Development Goals
Regulatory Bodies and Authorities	Payment Systems Department of Nepal Rastra Bank (NRB)	Licenses payment system operators and payment service providers (PSPs), sets payment system policies and standards, and oversees cross-border payment corridors and Central Bank Digital Currency pilots
	Financial Intelligence Unit (FIU-Nepal)	Serves as the national authority for anti-money laundering and combating the financing of terrorism oversight, and liaises with the Financial Action Task Force

continued on next page

Table A2 *continued*

Category	Stakeholder	Roles
Implementing Agencies	Nepal Telecommunications Authority	Regulates telecommunication operators, universal service obligations, and know-your-customer (KYC) process for subscriber identity module cards, and promotes last-mile connectivity to enable digital payments
	Nepal Bureau of Standards and Metrology (NBSM)	Issues national technical specifications for Quick Response (QR) code, ISO 20022 message profiles, and point-of-sale security
	Nepal Clearing House Limited (NCHL)	Operates the national payment infrastructure such as the National Payment Switch, real-time and batch automated clearing house systems, and National Payments Interface Exchange. It serves as the technical backbone for API-based trade payments
	Nepal National Single Window Secretariat	Coordinates over 40 border agencies via a unified portal
	Department of National Identification and Civil Registration	Issues biometric national identification cards, and provides digital identification verification and KYC services through APIs for licensed banks, financial institutions, and PSPs
Private Sector and Financial Partners	Commercial banks and PSPs (eSewa, Khalti, etc.)	Drive front-end digital payment innovation, merchant onboarding, and user adoption of QR and wallet-based payment
	Telecommunication operators (Nepal Telecom, Ncell)	Provides mobile connectivity and SMS and Unstructured Supplementary Service Data (USSD) channels, and supports nationwide QR payment acceptance
	NPCI International Payments Limited	Works with NRB and NCHL to enable QR code reciprocity and fast, low-cost Nepal-India payments
	Asian Development Bank and South Asia Subregional Economic Cooperation (SASEC)	Supply technical assistance and concessional finance for backbone upgrades, regulatory harmonization, and trade facilitation pilots SASEC brings together Bangladesh, Bhutan, India, Maldives, Myanmar, Nepal, and Sri Lanka to promote trade and cooperation

Source: Authors.

Advancing Digital Payments in Nepal

Infrastructure Upgrades and Policy Development for Enhanced Trade Facilitation

This report explores Nepal's evolving digital payment ecosystem as a driver of cross-border trade and economic growth and has been developed under the South Asia Subregional Economic Cooperation program with support from the Asian Development Bank. Aligned with Nepal's 16th Five-Year Plan and the Digital Nepal Framework, it highlights ongoing efforts to strengthen digital public infrastructure, including interoperable payment systems and digital identification. The report reviews existing developments and adoption trends, draws on relevant international experiences, and presents practical recommendations to support the continued development of secure, inclusive, and efficient digital payment systems that can facilitate Nepal's integration into regional and global markets.

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ASIAN DEVELOPMENT BANK

6 ADB Avenue, Mandaluyong City

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